NON-ACYCLICITY CLASSES AND A NON-ISOLATED MILNOR FORMULA

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To the memory of Professor Linsheng Yin (1963–2015)

Abstract. Let $f : X \to Y$ be a morphism between smooth schemes over a perfect field k. Let $\mathcal{F}$ be a constructible sheaf on X and Z a closed subscheme of X such that f is $SS(\mathcal{F})$-transversal outside Z. We construct a class supported on the non-transversality locus Z by using the characteristic cycle $CC(\mathcal{F})$ defined by T. Saito. This class is a geometric counterpart of the non-acyclicity class introduced by the second author and Zhao in [19]. Under certain conditions, the formation of this class is compatible with base change and proper pushforward. It also satisfies the Milnor formula proved by Saito and a conductor formula. We conjecture that the image of this class under the cycle class map is the non-acyclicity class. This conjecture can be viewed as a (relative version of) Milnor type formula for non-isolated singularities. We prove this conjecture when k is algebraically closed, X is quasi-affine and $Y = \mathbb{A}^1_k$. The proof uses a deformation to isolated characteristic points and compatibility with specialization.

Contents

1. Introduction

1.1. Let k be a perfect field of characteristic $p > 0$ and $S = \operatorname{Spec} k$. Let Λ be a finite field of characteristic $\ell \neq p$. Let X be a smooth scheme over S and $f : X \to Y$ a flat morphism of finite type to a smooth curve Y over S. If f has an isolated singularity at a closed point $x_0 \in |X|$, there is an invariant $\mu(X/Y, x_0)$ supported on x_0, called the Milnor number. The Milnor formula [7, Théorème 2.4] proved by Deligne says that the Milnor number is related to the total dimension at x_0 of the vanishing cycles $R\Phi(\Lambda, f)$ of f for the constant sheaf Λ, i.e.,

$$(-1)^n \mu(X/Y, x_0) = -\operatorname{dimtot} R\Phi_{\overline{x}_0}(\Lambda, f), \tag{1.1.1}$$

where $n = \dim X$ and $\operatorname{dimtot} = \dim + \mathrm{Sw}$ denotes the total dimension. Later in [9], Deligne proposed a Milnor formula for any constructible sheaf $\mathcal{F}$ of Λ-modules on X, which is realized and proved by Saito in [14]. If $x_0 \in |X|$ is at most an isolated characteristic point of f with respect to the singular support of $\mathcal{F}$, then Saito's theorem [14, Theorem 5.9] says

$$(CC(\mathcal{F}), df)_{T^*X, x_0} = -\operatorname{dimtot} R\Phi_{\overline{x}_0}(\mathcal{F}, f), \tag{1.1.2}$$

where $CC(\mathcal{F})$ is the characteristic cycle of $\mathcal{F}$. Now we propose the following question:

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Question 1.2. Is there a Milnor type formula for non-isolated singular/characteristic points?

1.3. If f is a projective flat morphism and if f is smooth outside $f^{-1}(y)$ for a closed point y of the curve Y, then the conductor formula of Bloch (cf. [15, Theorem 2.2.3 and Corollary 2.2.4])

$$-a_y(Rf_*\Lambda) = (-1)^n(X,X)_{T^*X,X_y} = (-1)^n \deg c^X_{n,X_y}(\Omega^1_{X/Y}) \cap [X] \tag{1.3.1}$$

gives a partial answer to Question 1.2. We view (1.1.1), (1.1.2) and (1.3.1) in the form

$$\deg(\text{geometric class on singular locus}) = \deg(\text{cohomology class on singular locus}). \tag{1.3.2}$$

We expect the equality holds without taking degree, i.e.,

$$\mathrm{cl}(\text{geometric class}) = \text{cohomology class}, \tag{1.3.3}$$

where cl is the cycle class map. In the paper [19], the third author and Yigeng Zhao introduced a (cohomological) non-acyclicity class which is supported on the non-acyclicity locus. Let Y be a smooth scheme over k and $X \to Y$ a separated morphism between schemes of finite type over k. Let $Z \subseteq X$ be a closed subscheme and $\mathcal{F} \in D_{\mathrm{ctf}}(X,\Lambda)$ such that $X \setminus Z \to Y$ is universally locally acyclic relatively to $\mathcal{F}|_{X\setminus Z}$. Then the cohomological non-acyclicity class $\widetilde{C}^Z_{X/Y/k}(\mathcal{F})$ is a class in $H^0_Z(X,\mathcal{K}_{X/Y/k})$, where $\mathcal{K}_{X/Y/k}$ sits in a distinguished triangle

$$\mathcal{K}_{X/Y} \to \mathcal{K}_{X/k} \to \mathcal{K}_{X/Y/k} \xrightarrow{+1}. \tag{1.3.4}$$

In this paper, we construct the geometric counterpart of the non-acyclicity class $\widetilde{C}^Z_{X/Y/k}(\mathcal{F})$. Precisely, when $X \to Y$ is a morphism between smooth schemes over k such that $X \to Y$ is $SS(\mathcal{F})$-transversal outside Z, then we construct a class $cc^Z_{X/Y/k}(\mathcal{F}) \in \mathrm{CH}_0(Z)$ (cf. (3.9.1)), called the geometric non-acyclicity class of $\mathcal{F}$. Explicitly, we put $r = \dim Y$. Let $q : \mathbb{P}(T^*Y \times_Y X) \to X$ be the projective bundle, with the tautological line bundle $\mathcal{O}(-1)$ over it. Let $\overline{df}$ be the section of $q^*T^*X \otimes \mathcal{O}(1)$ induced by $df : T^*Y \times_Y X \to T^*X$. Write $CC(\mathcal{F})_{\mathcal{O}(1)}$ for the conical cycle locally identified with $q^*CC(\mathcal{F})$ after trivializing $\mathcal{O}(1)$. Define

$$cc^Z_{X/Y/k}(\mathcal{F}) := q_{Z*}\left[c_{r-1}\left(\frac{q^*f^*T^*(Y/k)}{\mathcal{O}(-1)}\right) \cap \overline{df}^{\,!}CC(\mathcal{F})_{\mathcal{O}(1)}\right] \in \mathrm{CH}_0(Z). \tag{1.3.5}$$

There is a canonical extension $\overline{cc}_{X/Y}(\mathcal{F})$ of the relative characteristic class on $X\setminus Z$. For $\tau : Z \hookrightarrow X$, we have the fibration formula (3.12.1)

$$cc_{X/k}(\mathcal{F}) = c_{\mathrm{dimY}}(f^*\Omega^{1,\vee}_{Y/k}) \cap \overline{cc}_{X/Y}(\mathcal{F}) + \tau_* cc^Z_{X/Y/k}(\mathcal{F}). \tag{1.3.6}$$

Under certain conditions, we prove that the formation of the geometric non-acyclicity class $cc^Z_{X/Y/k}(\mathcal{F})$ is compatible with pullback (3.22.2) and proper pushforward (3.23.1). It also satisfies the Milnor formula (3.15.1) and a conductor formula (3.24.1). It is natural to expect the following conjecture holds:

Conjecture 1.4 (Conjecture 3.16)**.** *We have*

$$\widetilde{C}^Z_{X/Y/k}(\mathcal{F}) = \widetilde{\mathrm{cl}}(cc^Z_{X/Y/k}(\mathcal{F})) \quad \text{in} \quad H^0_Z(X,\mathcal{K}_{X/Y/k}), \tag{1.4.1}$$

where $\widetilde{\mathrm{cl}} : \mathrm{CH}_0(Z) \xrightarrow{\mathrm{cl}} H^0_Z(X,\mathcal{K}_{X/k}) \xrightarrow{(1.3.4)} H^0_Z(X,\mathcal{K}_{X/Y/k})$ *and* cl *is the cycle class map.*

We hope (1.4.1) gives a formulation of Question 1.2 in some sense.

In Section 4, we prove Conjecture 1.4 in the quasi-affine case for morphisms to the affine line over an algebraically closed field. More precisely, we have the following theorem.

Theorem 1.5 (Theorem 4.2)**.** *Let k be an algebraically closed field of characteristic $p>0$ and Λ a finite local ring such that the characteristic of its residue field is invertible in k. Let X be a smooth quasi-affine scheme of finite type over k, purely of dimension d. Let $f: X\to\mathbb{A}^1_k$ be a k-morphism and $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$. Put $Z=(df)^{-1}\,\mathrm{SS}(\mathcal{F})$, where $df: X\to T^*X$ is the section defined by the differential of f. Then we have*

$$\widetilde{C}^{Z}_{X/\mathbb{A}^1_k/k}(\mathcal{F})=\widetilde{\mathrm{cl}}(df^!\,\mathrm{CC}(\mathcal{F}))\quad \text{in } H^0_Z(X,\mathcal{K}_{X/\mathbb{A}^1_k/k}). \tag{1.5.1}$$

Here $df^!\,\mathrm{CC}(\mathcal{F})\in\mathrm{CH}_0(Z)$ is the refined Gysin pullback along the section df.

The proof proceeds by a deformation argument in the spirit of morsification in complex singularity theory. The quasi-affineness of X allows us to choose a global function g such that the generic member of the family $f+ug$, where u is the coordinate on the parameter line $\mathbb{A}^1_k$, has only isolated characteristic points (Proposition 4.13 and Lemma 4.15). After passing to the perfect closure of the generic point of $\mathbb{A}^1_k$, the desired equality follows from Saito's Milnor formula and the cohomological Milnor formula. We then descend this equality to the generic point and specialize to $u=0$. For this, we prove that the supported cycle class map is compatible with specialization (Theorem 4.8), and combine it with the specialization formulas for refined Gysin pullbacks and non-acyclicity classes.

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Notation and Conventions.

(1) Let S be a Noetherian scheme and Sch_S the category of separated schemes of finite type over S.
(2) Let Λ be a Noetherian ring such that $m\Lambda=0$ for some integer m invertible on S unless otherwise stated explicitly.
(3) For any scheme $X\in\mathrm{Sch}_S$, we denote by $D_{\mathrm{ctf}}(X,\Lambda)$ the derived category of complexes of Λ-modules of finite Tor-dimension with constructible cohomology sheaves on X.
(4) For any separated morphism $f: X\to Y$ in Sch_S, we use the following notation

$$\mathcal{K}_{X/Y}=Rf^!\Lambda,\quad D_{X/Y}(-)=R\mathcal{H}om(-,\mathcal{K}_{X/Y}). \tag{1.5.2}$$

(5) For $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$ and $\mathcal{G}\in D_{\mathrm{ctf}}(Y,\Lambda)$ on S-schemes X and Y respectively, $\mathcal{F}\boxtimes^L_S\mathcal{G}$ denotes $\mathrm{pr}_1^*\mathcal{F}\otimes^L\mathrm{pr}_2^*\mathcal{G}$ on $X\times_S Y$.
(6) To simplify our notation, we omit to write R or L to denote the derived functors unless otherwise stated explicitly or for $R\mathcal{H}om$.

2. Transversality condition

2.1. We recall the (cohomological) transversality condition introduced in [19, §2.1], which is a relative version of the transversality condition studied by Saito [14, Definition 8.5]. Consider the following Cartesian diagram in Sch_S:

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "p"'] \arrow[dr, phantom, "\square"] & Y \arrow[d, "f"] \\ W \arrow[r, "\delta"] & T. \end{tikzcd} \tag{2.1.1}$$

Let $\mathcal{F} \in D_{\mathrm{ctf}}(Y, \Lambda)$ and $\mathcal{G} \in D_{\mathrm{ctf}}(T, \Lambda)$. Let $c_{\delta,f,\mathcal{F},\mathcal{G}}$ be the composition

$$\begin{aligned} c_{\delta,f,\mathcal{F},\mathcal{G}} : i^*\mathcal{F} \otimes^L p^*\delta^!\mathcal{G} &\xrightarrow{id\otimes \mathrm{b.c}} i^*\mathcal{F} \otimes^L i^! f^*\mathcal{G} \\ &\xrightarrow{\mathrm{adj}} i^! i_!(i^*\mathcal{F} \otimes^L i^! f^*\mathcal{G}) \\ &\xrightarrow[\simeq]{\mathrm{proj.formula}} i^!(\mathcal{F} \otimes^L i_! i^! f^*\mathcal{G}) \xrightarrow{\mathrm{adj}} i^!(\mathcal{F} \otimes^L f^*\mathcal{G}). \end{aligned} \tag{2.1.2}$$

We put $c_{\delta,f,\mathcal{F}} := c_{\delta,f,\mathcal{F},\Lambda} : i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda \to i^!\mathcal{F}$. If $c_{\delta,f,\mathcal{F}}$ is an isomorphism, then we say that the morphism δ is *$\mathcal{F}$-transversal*. If $c_{i,\mathrm{id},\mathcal{F}}$ is an isomorphism, then we say i is $\mathcal{F}$-transversal.

By [19, §2.3], there is a functor $\delta^\Delta : D_{\mathrm{ctf}}(Y, \Lambda) \to D_{\mathrm{ctf}}(X, \Lambda)$ such that for any $\mathcal{F} \in D_{\mathrm{ctf}}(Y, \Lambda)$, we have a distinguished triangle

$$i^*\mathcal{F} \otimes^L p^*\delta^!\Lambda \xrightarrow{c_{\delta,f,\mathcal{F}}} i^!\mathcal{F} \to \delta^\Delta\mathcal{F} \xrightarrow{+1} . \tag{2.1.3}$$

Then δ is $\mathcal{F}$-transversal if and only if $\delta^\Delta(\mathcal{F}) = 0$ (cf. [19, Lemma 2.7]).

The following lemma gives an equivalence characterization between transversality condition and (universally) locally acyclicity condition.

Lemma 2.2. *Let $f : X \to S$ be a morphism separated of finite type between Noetherian schemes and $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. The following conditions are equivalent:*

(1) *The morphism f is locally acyclic relatively to $\mathcal{F}$.*
(2) *The morphism f is universally locally acyclic relatively to $\mathcal{F}$.*
(3) *For any $\mathcal{G} \in D_{\mathrm{ctf}}(X, \Lambda)$, the canonical map*

$$D_{X/S}(\mathcal{G}) \boxtimes^L_S \mathcal{F} \to R\mathcal{H}om_{X\times_S X}(\mathrm{pr}_1^*\mathcal{G}, \mathrm{pr}_2^!\mathcal{F}) \tag{2.2.1}$$

is an isomorphism in $D_{\mathrm{ctf}}(X \times_S X, \Lambda)$, where $\mathrm{pr}_1 : X \times_S X \to X$ and $\mathrm{pr}_2 : X \times_S X \to X$ are the projections.

(4) *The canonical map*

$$D_{X/S}(\mathcal{F}) \boxtimes^L_S \mathcal{F} \to R\mathcal{H}om_{X\times_S X}(\mathrm{pr}_1^*\mathcal{F}, \mathrm{pr}_2^!\mathcal{F}) \tag{2.2.2}$$

is an isomorphism.

(5) *For any Cartesian diagram*

$$\begin{tikzcd} Y \times_S X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\square"] & X \arrow[d, "f"] \\ Y \arrow[r, "\delta"'] & S \end{tikzcd} \tag{2.2.3}$$

such that all schemes are Noetherian and δ is separated of finite type, the morphism δ is $\mathcal{F}$-transversal.

(6) *For any Cartesian diagram (2.2.3) and any $\mathcal{G} \in D_{\mathrm{ctf}}(S, \Lambda)$, the morphism $c_{\delta,f,\mathcal{F},\mathcal{G}}$ is an isomorphism.*
(7) *For any Cartesian diagram*

$$\begin{tikzcd} Y \times_S X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] \arrow[dr, phantom, "\square"] & X' \arrow[r] \arrow[d, "f'"] \arrow[dr, phantom, "\square"] & X \arrow[d, "f"] \\ Y \arrow[r, "\delta"'] & S' \arrow[r] & S, \end{tikzcd} \tag{2.2.4}$$

such that all schemes are Noetherian and δ is separated of finite type, the morphism δ is $\mathcal{F}|_{X'}$-transversal.

(8) *For any Cartesian diagram* (2.2.4) *and any* $\mathcal{G}\in D_{\mathrm{ctf}}(S',\Lambda)$*, the morphism* $c_{\delta,f',\mathcal{F}|_{X'},\mathcal{G}}$ *is an isomorphism.*

When S is a scheme of finite type over a field k, then the equivalence between (2) and (7) follows from [19, Propositions 2.3(2) and 2.4]. In this case, we may require Y and S' smooth over k in (7).

Proof. By a result of Gabber [11, Corollary 6.6], (1) and (2) are equivalent. The equivalence between (2), (3), and (4) follows from [12, Proposition 2.5, Lemma 2.14, and Theorem 2.16]. By [19, Proposition 2.3(2)], (2) implies (6). It is clear that (6) implies (5).

Now we show (5) implies (1). By [12, Theorem 2.16], it suffices to show that for any $\mathcal{H}\in D_{\mathrm{ctf}}(Y,\Lambda)$, the canonical morphism

$$D_{Y/S}(\mathcal{H})\boxtimes^L_S\mathcal{F}\to R\mathcal{H}om(\mathrm{pr}_1^*\mathcal{H},\mathrm{pr}_2^!\mathcal{F}) \tag{2.2.5}$$

is an isomorphism. Using a standard dévissage argument [3, Exposé IX, Proposition 2.7], we may assume $\mathcal{H}=j_!\Lambda$ for $j:U\to Y$ étale with U affine. In this case, (2.2.5) is the following composition

$$j_*j^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}\to(j\times\mathrm{id})_*(j^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F})\xrightarrow[\simeq]{(2.2.8)}(j\times\mathrm{id})_*(j\times\mathrm{id})^*\mathrm{pr}_2^!\mathcal{F}. \tag{2.2.6}$$

In the following we show that

$$j_*j^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}\to(j\times\mathrm{id})_*(j^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \tag{2.2.7}$$

is an isomorphism. We write j as a composition of a proper morphism and an open immersion.

Since δ is $\mathcal{F}$-transversal, we have an isomorphism

$$\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}\xrightarrow{\simeq}\mathrm{pr}_2^!\mathcal{F}. \tag{2.2.8}$$

By the projection formula and proper base change, for any proper morphism $g:Y'\to Y$ and $A\in D_{\mathrm{ctf}}(Y',\Lambda)$, the following canonical morphism

$$g_!A\boxtimes^L_S\mathcal{F}\xrightarrow{\simeq}(g\times\mathrm{id})_!(A\boxtimes^L_S\mathcal{F}) \tag{2.2.9}$$

is an isomorphism. If $u:Y'\to Y$ is an open immersion with closed complementary $\tau:Z=Y\backslash Y'\to Y$, we have a commutative diagram between distinguished triangles

(2.2.10)

$$\begin{tikzcd}
\tau_!\tau^!\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F} \arrow[r] \arrow[d, "(2.2.9)"', "\simeq"] & \mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F} \arrow[r] \arrow[d, "\simeq"] & u_*u^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F} \arrow[r, "+1"] \arrow[d] & {} \\
(\tau\times\mathrm{id})_!(\tau^!\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \arrow[r] \arrow[d, "(2.2.8)"', "\simeq"] & \mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F} \arrow[r] \arrow[d, equal] & (u\times\mathrm{id})_*(u^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \arrow[r, "+1"] \arrow[d, "\simeq"] & {} \\
(\tau\times\mathrm{id})_!(\tau\times\mathrm{id})^!(\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \arrow[r] & \mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F} \arrow[r] & (u\times\mathrm{id})_*(u\times\mathrm{id})^*(\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \arrow[r, "+1"] & {}.
\end{tikzcd}$$

Thus

$$u_*u^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}\to(u\times\mathrm{id})_*(u^*\mathcal{K}_{Y/S}\boxtimes^L_S\mathcal{F}) \tag{2.2.11}$$

is an isomorphism. Applying (2.2.11) and (2.2.9), we obtain that (2.2.7) is an isomorphism, and the conclusion follows.

We show (2) implies (8). By assumption, f' is also universally locally acyclic relatively to $\mathcal{F}|_{X'}$. Thus by (6), the morphism $c_{\delta,f',\mathcal{F}|_{X'},\mathcal{G}}$ is an isomorphism.

Finally, it is clear that (8) implies (7) and (7) implies (5). □

2.3. Now we recall the geometric transversality condition (cf. [4, 1.2] and [14, Definitions 3.3, 3.5, and 7.1]). Let X be a smooth scheme over a field k. Let C be a conical closed subset of T^*X, i.e., a closed subset which is stable under the action of the multiplicative group $\mathbb{G}_m$. We denote by $T^*_X X \subseteq T^*X$ the zero section of the cotangent bundle T^*X of X.

(1) Let $h\colon W \to X$ be a morphism from a smooth scheme W over k. We say that h is C-transversal if the intersection $(C \times_X W) \cap dh^{-1}(T^*_W W)$ is contained in the zero-section $T^*_X X \times_X W \subseteq T^*X \times_X W$, where $dh\colon T^*X \times_X W \to T^*W$ is the canonical map.

(2) Assume that X and C are purely of dimension d and that W is purely of dimension m. We say that a C-transversal map $h\colon W \to X$ is properly C-transversal if every irreducible component of $C \times_X W$ is of dimension m.

(3) We say that a morphism $f\colon X \to Y$ to a smooth scheme Y over k is C-transversal if the inverse image $df^{-1}(C)$ is contained in the zero-section $T^*_Y Y \times_Y X \subseteq T^*Y \times_Y X$, where $df\colon T^*Y \times_Y X \to T^*X$ is the canonical map.

The cohomological transversality condition and geometric transversality condition are related as follows. Let X be a smooth scheme of pure dimension d and Λ a finite local ring such that the characteristic ℓ of the residue field of Λ is invertible in k. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X,\Lambda)$. The singular support $SS(\mathcal{F})$ defined by Beilinson [4] is a d-dimensional conical closed subset of T^*X. We have the following properties:

(1) $SS(\mathcal{F}) \cap T^*_X X = \mathrm{supp}(\mathcal{F})$.

(2) Let $f : X \to Y$ be a morphism to a smooth scheme Y over k. If f is $SS(\mathcal{F})$-transversal, then f is universally locally acyclic relatively to $\mathcal{F}$.

(3) Let $f : W \to X$ be a morphism from a smooth scheme W over k. If f is $SS(\mathcal{F})$-transversal, then f is $\mathcal{F}$-transversal.

Now assume k is a perfect field. By [14, Theorem 5.9 and Theorem 5.18], the characteristic cycle $CC(\mathcal{F})$ is the unique d-cycle $CC(\mathcal{F})$ supported on $SS(\mathcal{F})$ with $\mathbb{Z}$-coefficients such that $CC(\mathcal{F})$ satisfies the Milnor formula (1.1.2) for $\mathcal{F}$.

3. Geometric non-acyclicity classes

3.1. Let S be a Noetherian scheme and Λ a Noetherian ring such that $m\Lambda = 0$ for some integer m invertible on S. Consider the following Cartesian diagram in Sch_S

$$\begin{tikzcd} X \times_S Y \arrow[r, "\mathrm{pr}_1"] \arrow[d, "\mathrm{pr}_2"'] \arrow[dr, phantom, "\Box"] & X \arrow[d, "h"] \\ Y \arrow[r, "g"] & S, \end{tikzcd} \tag{3.1.1}$$

where pr_1 and pr_2 are the projections. For any $\mathcal{F} \in D_{\mathrm{ctf}}(X,\Lambda)$ and $\mathcal{G} \in D_{\mathrm{ctf}}(Y,\Lambda)$, we have canonical morphisms

$$\mathcal{F} \boxtimes^L_S \mathcal{K}_{Y/S} = \mathrm{pr}_1^*\mathcal{F} \otimes^L \mathrm{pr}_2^* g^!\Lambda \xrightarrow{c_{g,h,\mathcal{F}}} \mathrm{pr}_1^!\mathcal{F}, \tag{3.1.2}$$

$$\mathcal{F} \boxtimes^L_S D_{Y/S}(\mathcal{G}) \to R\mathcal{H}om_{X\times_S Y}(\mathrm{pr}_2^*\mathcal{G}, \mathrm{pr}_1^!\mathcal{F}), \tag{3.1.3}$$

where (3.1.3) is adjoint to

$$\mathcal{F} \boxtimes^L_S (D_{Y/S}(\mathcal{G}) \otimes^L \mathcal{G}) \xrightarrow{id\boxtimes \mathrm{ev}} \mathcal{F} \boxtimes^L_S \mathcal{K}_{Y/S} \xrightarrow{(3.1.2)} \mathrm{pr}_1^!\mathcal{F}. \tag{3.1.4}$$

Note that (3.1.2) is a special case of (3.1.3) by taking $\mathcal{G} = \Lambda$. If moreover $X \to S$ is universally locally acyclic relatively to $\mathcal{F}$, then (3.1.3) is an isomorphism by (2.2.5). For a morphism $c =$

$(c_1, c_2) : C \to X \times_S Y$, we have a canonical isomorphism by [6, Corollaire 3.1.12.2]

(3.1.5) $$R\mathcal{H}om(c_2^*\mathcal{G}, c_1^!\mathcal{F}) \xrightarrow{\simeq} c^! R\mathcal{H}om(\mathrm{pr}_2^*\mathcal{G}, \mathrm{pr}_1^!\mathcal{F}).$$

3.2. Consider a commutative diagram in Sch_S:

(3.2.1) $$\begin{tikzcd} Z \arrow[r, hook, "\tau"] & X \arrow[rr, "f"] \arrow[dr, "h"'] & & Y, \arrow[dl, "g"] \\ & & S & \end{tikzcd}$$

where $\tau : Z \to X$ is a closed immersion and g is a smooth morphism. Let $i : X \times_Y X \to X \times_S X$ be the base change of the diagonal morphism $\delta : Y \to Y \times_S Y$:

(3.2.2) $$\begin{tikzcd} X \arrow[r, equal] \arrow[d, hook, "\delta_1"] \arrow[dd, bend right=60, "f"'] \arrow[dr, phantom, "\square"] & X \arrow[d, hook, "\delta_0"] \\ X \times_Y X \arrow[r, "i"] \arrow[d, "p"] \arrow[dr, phantom, "\square"] & X \times_S X \arrow[d, "f \times f"] \\ Y \arrow[r, "\delta"] & Y \times_S Y, \end{tikzcd}$$

where δ_0 and δ_1 are the diagonal morphisms. Put $\mathcal{K}_{X/Y/S} := \delta^\Delta \mathcal{K}_{X/S} \simeq \delta_1^* \delta^\Delta \delta_{0*} \mathcal{K}_{X/S}$. By (2.1.3), we have the following distinguished triangle (see also [19, (4.6)])

(3.2.3) $$\mathcal{K}_{X/Y} \to \mathcal{K}_{X/S} \to \mathcal{K}_{X/Y/S} \xrightarrow{+1}.$$

Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that $X \setminus Z \to Y$ is universally locally acyclic relatively to $\mathcal{F}|_{X \setminus Z}$ and that $h : X \to S$ is universally locally acyclic relatively to $\mathcal{F}$. We put

(3.2.4) $$\mathcal{H}_S = R\mathcal{H}om_{X \times_S X}(\mathrm{pr}_2^*\mathcal{F}, \mathrm{pr}_1^!\mathcal{F}), \qquad \mathcal{T}_S = \mathcal{F} \boxtimes_S^L D_{X/S}(\mathcal{F}).$$

The relative cohomological characteristic class $C_{X/S}(\mathcal{F})$ is the composition (cf. [19, (3.49)])

(3.2.5) $$\Lambda \xrightarrow{\mathrm{id}} R\mathcal{H}om(\mathcal{F}, \mathcal{F}) \xrightarrow[\simeq]{(3.1.5)} \delta_0^! \mathcal{H}_S \xleftarrow[\simeq]{(3.1.3)} \delta_0^! \mathcal{T}_S \to \delta_0^* \mathcal{T}_S \xrightarrow{\mathrm{ev}} \mathcal{K}_{X/S}.$$

By the assumption on $\mathcal{F}$, $\delta_1^* \delta^\Delta \mathcal{T}_S$ is supported on Z by [19, §4.3]. The non-acyclicity class $\widetilde{C}^Z_{X/Y/S}(\mathcal{F})$ is the composition (cf. [19, Definition 4.2])

(3.2.6) $$\Lambda \to \delta_0^! \mathcal{H}_S \xleftarrow{\simeq} \delta_0^! \mathcal{T}_S \simeq \delta_1^! i^! \mathcal{T}_S \to \delta_1^* i^! \mathcal{T}_S \to \delta_1^* \delta^\Delta \mathcal{T}_S \xleftarrow{\simeq} \tau_* \tau^! \delta_1^* \delta^\Delta \mathcal{T}_S \to \tau_* \tau^! \mathcal{K}_{X/Y/S}.$$

If the following condition holds:

(3.2.7) $$H^0(Z, \mathcal{K}_{Z/Y}) = 0 \text{ and } H^1(Z, \mathcal{K}_{Z/Y}) = 0$$

then the map

(3.2.8) $$H^0_Z(X, \mathcal{K}_{X/S}) \xrightarrow{(3.2.3)} H^0_Z(X, \mathcal{K}_{X/Y/S})$$

is an isomorphism. In this case, the class $\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) \in H^0_Z(X, \mathcal{K}_{X/Y/S})$ defines an element of $H^0_Z(X, \mathcal{K}_{X/S})$, which is denoted by $C^Z_{X/Y/S}(\mathcal{F})$. Now we summarize the functorial properties for the non-acyclicity classes.

Proposition 3.3 (Yang-Zhao[19, Theorem 1.5, Proposition 1.6, Theorem 1.7, and Theorem 1.8])**.** *Let us denote the diagram* (3.2.1) *by* $\Delta = \Delta^Z_{X/Y/S}$ *and* $\widetilde{C}^Z_{X/Y/S}(\mathcal{F})$, $C^Z_{X/Y/S}(\mathcal{F})$ *by* $\widetilde{C}_\Delta(\mathcal{F})$, $C_\Delta(\mathcal{F})$, *respectively. Let* $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. *Assume that* $Y \to S$ *is smooth of pure relative dimension* r, $X \setminus Z \to Y$ *is universally locally acyclic relatively to* $\mathcal{F}|_{X\setminus Z}$ *and that* $X \to S$ *is universally locally acyclic relatively to* $\mathcal{F}$.

(1) (*Fibration formula*) *If* $H^0(Z, \mathcal{K}_{Z/Y}) = H^1(Z, \mathcal{K}_{Z/Y}) = 0$, *then we have*

$$C_{X/S}(\mathcal{F}) = c_r(f^*\Omega^{1,\vee}_{Y/S}) \cap C_{X/Y}(\mathcal{F}) + C_\Delta(\mathcal{F}) \quad \text{in} \quad H^0(X, \mathcal{K}_{X/S}). \tag{3.3.1}$$

(2) (*Pullback*) *Let* $b : S' \to S$ *be a morphism of Noetherian schemes. Let* $\Delta' = \Delta^{Z'}_{X'/Y'/S'}$ *be the base change of* $\Delta = \Delta^Z_{X/Y/S}$ *by* $b : S' \to S$. *Let* $b_X : X' = X \times_S S' \to X$ *be the base change of* b *by* $X \to S$. *Then we have*

$$b_X^* \widetilde{C}_\Delta(\mathcal{F}) = \widetilde{C}_{\Delta'}(b_X^*\mathcal{F}) \quad \text{in} \quad H^0_{Z'}(X', \mathcal{K}_{X'/Y'/S'}), \tag{3.3.2}$$

where $b_X^* : H^0_Z(X, \mathcal{K}_{X/Y/S}) \to H^0_{Z'}(X', \mathcal{K}_{X'/Y'/S'})$ *is the induced pullback morphism.*

(3) (*Proper pushforward*) *Consider a diagram* $\Delta' = \Delta^{Z'}_{X'/Y/S}$. *Let* $s : X \to X'$ *be a proper morphism over* Y *such that* $Z \subseteq s^{-1}(Z')$. *Then we have*

$$s_*(\widetilde{C}_\Delta(\mathcal{F})) = \widetilde{C}_{\Delta'}(Rs_*\mathcal{F}) \quad \text{in} \quad H^0_{Z'}(X', \mathcal{K}_{X'/Y/S}), \tag{3.3.3}$$

where $s_* : H^0_Z(X, \mathcal{K}_{X/Y/S}) \to H^0_{Z'}(X', \mathcal{K}_{X'/Y/S})$ *is the induced pushforward morphism.*

(4) (*Cohomological Milnor formula*) *Assume* $S = \mathrm{Spec}\, k$ *for a perfect field* k *of characteristic* $p > 0$ *and* Λ *is a finite local ring such that the characteristic of the residue field is invertible in* k. *If* $Z = \{x\}$, *then we have*

$$C_\Delta(\mathcal{F}) = -\,\mathrm{dimtot}\, R\Phi_{\bar{x}}(\mathcal{F}, f) \quad \text{in} \quad \Lambda = H^0_x(X, \mathcal{K}_{X/k}), \tag{3.3.4}$$

where $R\Phi(\mathcal{F}, f)$ *is the complex of vanishing cycles and* $\mathrm{dimtot} = \dim + \mathrm{Sw}$ *is the total dimension.*

(5) (*Cohomological conductor formula*) *Assume* $S = \mathrm{Spec}\, k$ *for a perfect field* k *of characteristic* $p > 0$ *and* Λ *is a finite local ring such that the characteristic of the residue field is invertible in* k. *If* Y *is a smooth connected curve over* k, f *is proper and* $Z = f^{-1}(y)$ *for a closed point* $y \in |Y|$, *then we have*

$$f_*\widetilde{C}_\Delta(\mathcal{F}) = -a_y(Rf_*\mathcal{F}) \quad \text{in} \quad \Lambda = H^0_y(Y, \mathcal{K}_{Y/k}), \tag{3.3.5}$$

where $a_y(\mathcal{G}) = \mathrm{rank}\mathcal{G}|_{\bar{\eta}} - \mathrm{rank}\mathcal{G}_{\bar{y}} + \mathrm{Sw}_y\mathcal{G}$ *is the Artin conductor of an object* $\mathcal{G} \in D_{\mathrm{ctf}}(Y, \Lambda)$ *at* y *and* η *is the generic point of* Y.

(6) (*Specialization;* [19, Proposition 4.7]) *Assume that* S *is a henselian trait with generic point* η, *and let* s *be a geometric point above its closed point. Put* $\Delta_t = \Delta^{Z_t}_{X_t/Y_t/t}$ *and* $\mathcal{F}_t = \mathcal{F}|_{X_t}$ *for* $t \in \{\eta, s\}$. *For this item, the hypothesis on* f *may be weakened to universal local acyclicity of* f_t *relatively to* $\mathcal{F}_t$ *outside* Z_t. *Then we have*

$$\mathrm{sp}^{\text{ét}}(\widetilde{C}_{\Delta_\eta}(\mathcal{F}_\eta)) = \widetilde{C}_{\Delta_s}(\mathcal{F}_s) \quad \textit{in } H^0_{Z_s}(X_s, K_{X_s/Y_s/s}), \tag{3.3.6}$$

where

$$\mathrm{sp}^{\text{ét}} : H^0_{Z_\eta}(X_\eta, K_{X_\eta/Y_\eta/\eta}) \longrightarrow H^0_{Z_s}(X_s, K_{X_s/Y_s/s}) \tag{3.3.7}$$

is the specialization map

3.4. Let X be a smooth connected curve over k. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ and $Z \subseteq X$ be a finite set of closed points such that the cohomology sheaves of $\mathcal{F}|_{X \setminus Z}$ are locally constant. By the cohomological Milnor formula (3.3.4), we have the following (motivic) expression for the Artin conductor of $\mathcal{F}$ at $x \in Z$

$$a_x(\mathcal{F}) = \mathrm{dimtot}\, R\Phi_{\bar{x}}(\mathcal{F}, \mathrm{id}) = -C^{\{x\}}_{U/U/k}(\mathcal{F}|_U), \tag{3.4.1}$$

where U is any open subscheme of X such that $U \cap Z = \{x\}$. By (3.3.1), we get the following cohomological Grothendieck-Ogg-Shafarevich formula (cf. [19, Corollary 6.4]):

$$C_{X/k}(\mathcal{F}) = \mathrm{rank}\mathcal{F} \cdot c_1(\Omega^{1,\vee}_{X/k}) - \sum_{x \in Z} a_x(\mathcal{F}) \cdot [x] \quad \text{in} \quad H^0(X, \mathcal{K}_{X/k}). \tag{3.4.2}$$

3.5. Now we start to construct and prove a geometric counterpart of Proposition 3.3. Let k be a perfect field of characteristic $p > 0$ and Λ be a finite local ring whose residue field is of characteristic $\ell \neq p$. Let Sm_k be the category of smooth schemes over k. Let S be a smooth connected scheme of dimension s over k. Let $f : X \to S$ be a morphism in Sm_k. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that f is $SS(\mathcal{F})$-transversal. Consider the following morphisms

$$X \xrightarrow{0} T^*S \times_S X \xrightarrow{df} T^*X, \tag{3.5.1}$$

where 0 stands for the zero section. By assumption, $df^{-1}(SS(\mathcal{F}))$ is contained in $0(X)$. We define the relative characteristic class of $\mathcal{F}$ to be the following s-cycle class on X:

$$cc_{X/S}(\mathcal{F}) := (-1)^s \cdot (df)^!(CC(\mathcal{F})) \quad \text{in} \quad \mathrm{CH}_s(X), \tag{3.5.2}$$

where $(df)^!$ is the refined Gysin pullback. When f is a smooth morphism, then we have a Cartesian diagram

$$\begin{tikzcd} T^*S \times_S X \arrow[r, "df"] \arrow[d] \arrow[dr, phantom, "\square"] & T^*X \arrow[d] \\ X \arrow[r, "0_{X/S}"'] & T^*(X/S). \end{tikzcd} \tag{3.5.3}$$

In this case, we have $cc_{X/S}(\mathcal{F}) = (-1)^s \cdot 0^!_{X/S}(CC(\mathcal{F}))$ (cf. [18, Definition 2.11]). If f is a smooth morphism of relative dimension r and if $\mathcal{F}$ is locally constant, then we have

$$cc_{X/S}(\mathcal{F}) = (-1)^s \cdot 0^!_{X/S}((-1)^{\dim X} \cdot \mathrm{rank}\mathcal{F} \cdot [T^*_X X]) = \mathrm{rank}\mathcal{F} \cdot c_r(\Omega^{1,\vee}_{X/S}) \cap [X]. \tag{3.5.4}$$

We have the following conjecture, c.f. [18, Conjecture 3.8]:

Conjecture 3.6. *Let S be a smooth connected scheme of dimension s over k. Let $f : X \to S$ be a morphism in Sm_k. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that f is $SS(\mathcal{F})$-transversal. Then we have*

$$\mathrm{cl}(cc_{X/S}(\mathcal{F})) = C_{X/S}(\mathcal{F}) \quad \text{in} \quad H^0(X, \mathcal{K}_{X/S}), \tag{3.6.1}$$

where $\mathrm{cl} : \mathrm{CH}_s(X) \to H^0(X, \mathcal{K}_{X/S})$ *is the cycle class map.*

When $S = \mathrm{Spec}\, k$, then it is Saito's conjecture [14, Conjecture 6.8.1], which is proved under quasi-projective assumption in [19, Theorem 1.2]. When $f : X \to S$ is a smooth morphism, then (3.6.1) is true for a locally constant constructible (flat) sheaf $\mathcal{F}$ of Λ-modules. Indeed, this follows from (3.5.4), [19, Lemma 3.1] and (3.3.1).

Remark 3.7. By [10, Chapter 17], $\mathrm{CH}_s(X) \cong \mathrm{CH}^0(X \to S)$ where the latter bivariant Chow group can be regarded as a relative version of $\mathrm{CH}_0(X)$. However, in general we cannot expect $cc_{X/S}(\mathcal{F})$ to lift canonically to a relative cycle.

3.8. Consider a commutative diagram

$$\begin{tikzcd} Z \arrow[r, hook, "\tau"] & X \arrow[rr, "f"] \arrow[dr, "h"'] & & Y \arrow[dl, "g"] \\ & & S & \end{tikzcd} \tag{3.8.1}$$

where X, Y, S belong to Sm_k and $\tau : Z \hookrightarrow X$ is a closed immersion. Assume that X, Y, S are of pure dimension, that S is connected of dimension s, and that g is smooth of relative dimension $r > 0$. Put $n = \dim X$. Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that h is $SS(\mathcal{F})$-transversal and $f|_{X\setminus Z}$ is $SS(\mathcal{F}|_{X\setminus Z})$-transversal. We use the convention that $\mathbb{P}$ parametrizes lines. Write

$$q : \mathbb{P}(T^*Y \times_Y X) \longrightarrow X. \tag{3.8.2}$$

The tautological subbundle and the differential give

$$\mathcal{O}(-1) \hookrightarrow q^*(T^*Y \times_Y X) \xrightarrow{q^*df} q^*T^*X. \tag{3.8.3}$$

Under $\mathcal{H}om(\mathcal{O}(-1), q^*T^*X) = q^*T^*X \otimes \mathcal{O}(1)$, this homomorphism defines a section of the vector bundle $q^*T^*X \otimes \mathcal{O}(1)$:

$$\overline{df} : \mathbb{P}(T^*Y \times_Y X) \longrightarrow q^*T^*X \otimes \mathcal{O}(1). \tag{3.8.4}$$

At a geometric point (x, λ), with $\lambda \subset T^*_{f(x)}Y$ a line, its value is the map $df_x|_\lambda : \lambda \to T^*_x X$. Thus $\overline{df}$ is defined even when $df_x(\lambda) = 0$. It is a section of a vector bundle.

For a conical closed subscheme $D \subset T^*X$, we define its twist. On an open $V \subset \mathbb{P}(T^*Y \times_Y X)$ with local basis e of $\mathcal{O}(1)$, transport $D \times_X V$ by

$$q^*T^*X|_V \xrightarrow{\sim} (q^*T^*X \otimes \mathcal{O}(1))|_V, \qquad \xi \longmapsto \xi \otimes e. \tag{3.8.5}$$

Changing e multiplies the vector coordinate by a unit. Since the ideal of D is homogeneous, these closed subschemes agree on overlaps and define $D_{\mathcal{O}(1)} \subset q^*T^*X \otimes \mathcal{O}(1)$. Equivalently, if $D = \mathrm{Spec}_X(\bigoplus_{a\geq 0} \mathcal{A}_a)$ with its graded coordinate algebra, then

$$D_{\mathcal{O}(1)} = \mathrm{Spec}_{\mathbb{P}(T^*Y\times_Y X)} \left(\bigoplus_{a\geq 0} q^*\mathcal{A}_a \otimes \mathcal{O}(-a) \right). \tag{3.8.6}$$

For $CC(\mathcal{F}) = \sum_a m_a[C_a]$, set

$$CC(\mathcal{F})_{\mathcal{O}(1)} := \sum_a m_a[(C_a)_{\mathcal{O}(1)}]. \tag{3.8.7}$$

Each $(C_a)_{\mathcal{O}(1)}$ is integral of dimension $n + r + s - 1$. We apply the same twist separately to $SS(\mathcal{F})$.

Notice that we have the following Cartesian diagram

$$\begin{tikzcd} \mathbb{P}((df)^{-1}SS(\mathcal{F})) \arrow[r] \arrow[d] & SS(\mathcal{F})_{\mathcal{O}(1)} \arrow[d] \\ \mathbb{P}(T^*Y \times_Y X) \arrow[r, "\overline{df}"'] & q^*T^*X \otimes \mathcal{O}(1) \end{tikzcd} \tag{3.8.8}$$

which can be checked on open subschemes. Moreover, the section $\overline{df}$ is a regular embedding of codimension n, so

$$\overline{df}^!CC(\mathcal{F})_{\mathcal{O}(1)} \in \mathrm{CH}_{r+s-1}(\mathbb{P}((df)^{-1}SS(\mathcal{F}))). \tag{3.8.9}$$

The hypotheses imply

$$(df)^{-1}SS(\mathcal{F}) \subset 0(X) \cup (T^*Y \times_Y Z), \qquad (df)^{-1}SS(\mathcal{F}) \cap (T^*S \times_S X) \subset 0(X). \tag{3.8.10}$$

Consequently, on $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$, the tautological line bundle $\mathcal{O}(-1)$ is a subbundle of $q^*f^*T^*(Y/S)$, by the short exact sequence of bundles

$$0 \to T^*S \times_S X \to T^*Y \times_Y X \to f^*T^*(Y/S) \to 0. \tag{3.8.11}$$

The reduced closed subscheme of the support $\mathbb{P}\left((df)^{-1}SS(\mathcal{F})\right)$ maps properly to Z_{red} and hence to Z. For this pushforward, denoted by q_{Z*}, we use the canonical invariance of Chow groups under reduction. The refined intersection retains its scheme structure and intersection multiplicities.

Definition 3.9. We define the geometric non-acyclicity class of $\mathcal{F}$ by

$$cc^Z_{X/Y/S}(\mathcal{F}) := (-1)^s q_{Z*}\left[c_{r-1}\left(\frac{q^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right) \cap \overline{df}^!CC(\mathcal{F})_{\mathcal{O}(1)}\right] \in \mathrm{CH}_s(Z). \tag{3.9.1}$$

Here, the quotient bundle is taken on $\mathbb{P}((df)^{-1}SS(\mathcal{F}))$.

Proposition 3.10. *Consider the previous assumptions.*

(1) *If f is $SS(\mathcal{F})$-transversal on X, then $cc^Z_{X/Y/S}(\mathcal{F}) = 0$ for any Z;*

(2) *Let $Z \subset Z_1 \subset X$ be closed subschemes, with inclusion $i : Z \hookrightarrow Z_1$. Then*

$$i_*cc^Z_{X/Y/S}(\mathcal{F}) = cc^{Z_1}_{X/Y/S}(\mathcal{F}); \tag{3.10.1}$$

(3) *Let*

$$\mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \xrightarrow{+1} \tag{3.10.2}$$

be a distinguished triangle in $D_{\mathrm{ctf}}(X, \Lambda)$, then we have

$$cc^Z_{X/Y/S}(\mathcal{F}) = cc^Z_{X/Y/S}(\mathcal{F}') + cc^Z_{X/Y/S}(\mathcal{F}''). \tag{3.10.3}$$

Proof. The parts (1) and (2) are direct from definition. For (3), use [14, Lemma 5.13]. □

3.11. For a vector bundle V of rank d on X, put

$$c_t(V) = t^d + c_1(V)t^{d-1} + \cdots + c_d(V) \in \mathrm{CH}^*(X)[t]. \tag{3.11.1}$$

We regard t as a formal variable of codimension one. Recall that

$$\frac{1}{c_t(V)} = t^{-d}\sum_{a\geq 0} s_a(V)t^{-a}, \tag{3.11.2}$$

where $s(V) = c(V)^{-1}$ is the total Segre class.

Let $n = \dim X$, and write

$$0_X : X \longrightarrow T^*X, \qquad q_X : \mathbb{P}(T^*X) \longrightarrow X \tag{3.11.3}$$

for the zero section and the projection. Put $\xi_X = c_1(\mathcal{O}_{\mathbb{P}(T^*X)}(1))$. Write

$$CC(\mathcal{F}) = 0_{X*}\Gamma + \sum_a m_a[C_a], \tag{3.11.4}$$

where Γ is an n-cycle on X and no C_a is contained in the zero section. Since each C_a is conical, $\mathbb{P}(C_a)$ is a closed integral subscheme of $\mathbb{P}(T^*X)$ of dimension $n-1$.

By the projective bundle formula, there are unique classes

$$\beta_j \in \mathrm{CH}_j(X), \qquad 0 \leq j \leq n-1, \tag{3.11.5}$$

such that

$$\sum_a m_a[\mathbb{P}(C_a)] = \sum_{j=0}^{n-1} \xi_X^j \cap q_X^* \beta_j \quad \text{in } \mathrm{CH}_{n-1}(\mathbb{P}(T^*X)). \tag{3.11.6}$$

Define $b_j \in \mathrm{CH}_j(X)$ by

$$\sum_{j=0}^{n} b_j t^j = c_t(T^*X) \cap \Gamma + \sum_{j=0}^{n-1} \beta_j t^j, \tag{3.11.7}$$

where

$$c_t(T^*X) \cap \Gamma = \sum_{i=0}^{n} (c_i(T^*X) \cap \Gamma) t^{n-i}. \tag{3.11.8}$$

In particular, $b_n = \Gamma$. The classes b_j are canonically determined by $CC(\mathcal{F})$. Indeed, by the projective bundle relation

$$\xi_X^n + c_1(T^*X)\xi_X^{n-1} + \cdots + c_n(T^*X) = 0 \tag{3.11.9}$$

we have

$$\sum_{j=0}^{n} \xi_X^j \cap q_X^* b_j = \sum_a m_a[\mathbb{P}(C_a)]. \tag{3.11.10}$$

Put $m = \dim Y = r + s$. We define

$$\overline{cc}_{X/Y}(\mathcal{F}) := (-1)^m \sum_{j \geq m} s_{j-m}(T^*Y \times_Y X) \cap b_j \in \mathrm{CH}_m(X). \tag{3.11.11}$$

Equivalently, $\overline{cc}_{X/Y}(\mathcal{F})$ is the constant coefficient of the Laurent expansion at $t = \infty$ of

$$(-1)^m (\sum_{j=0}^{n} b_j t^j) \cdot c_t(T^*Y \times_Y X)^{-1}. \tag{3.11.12}$$

Proposition 3.12 (fibration formula)**.** *Consider the previous assumptions, then the restriction of $\overline{cc}_{X/Y}(\mathcal{F})$ to $X \setminus Z$ is $cc_{(X\setminus Z)/Y}(\mathcal{F}|_{X\setminus Z})$. Moreover,*

$$cc_{X/S}(\mathcal{F}) = c_r((f^*\Omega^1_{Y/S})^\vee) \cap \overline{cc}_{X/Y}(\mathcal{F}) + \tau_* cc^Z_{X/Y/S}(\mathcal{F}) \quad \textit{in } \mathrm{CH}_s(X). \tag{3.12.1}$$

In particular, if $\dim Z < r + s$, the restriction

$$\mathrm{CH}_{r+s}(X) \longrightarrow \mathrm{CH}_{r+s}(X \setminus Z) \tag{3.12.2}$$

is an isomorphism, and $\overline{cc}_{X/Y}(\mathcal{F})$ is the unique extension of $cc_{(X\setminus Z)/Y}(\mathcal{F}|_{X\setminus Z})$. If $Z = \varnothing$, then

$$cc_{X/S}(\mathcal{F}) = c_r((f^*\Omega^1_{Y/S})^\vee) \cap cc_{X/Y}(\mathcal{F}). \tag{3.12.3}$$

Proof. Since $h : X \to S$ is $SS(\mathcal{F})$-transversal, the inverse image of $SS(\mathcal{F})$ under

$$dh : T^*S \times_S X \longrightarrow T^*X \tag{3.12.4}$$

is contained in the zero section, hence every projectivized nonzero component of $CC(\mathcal{F})$ is disjoint from $\mathbb{P}(T^*S \times_S X)$. Hence (3.11.10) restricts to

$$\sum_j \xi^j \cap q^* b_j = 0 \tag{3.12.5}$$

on $\mathbb{P}(T^*S \times_S X)$. By the projective bundle formula, we have $\sum_j b_j t^j$ is divisible by $c_t(T^*S \times_S X)$. Write

$$\sum_{j=0}^{n} b_j t^j = c_t(T^*S \times_S X) \sum_{j\geq 0} a_j t^j, \qquad a_j \in \mathrm{CH}_{s+j}(X). \tag{3.12.6}$$

The constant coefficient is the ordinary refined intersection with dh:

$$a_0 = (dh)^! CC(\mathcal{F}), \qquad cc_{X/S}(\mathcal{F}) = (-1)^s a_0. \tag{3.12.7}$$

Since $g : Y \to S$ is smooth of relative dimension r, there is an exact sequence of vector bundles on X

$$0 \longrightarrow T^*S \times_S X \longrightarrow T^*Y \times_Y X \longrightarrow f^*T^*(Y/S) \longrightarrow 0. \tag{3.12.8}$$

Hence

$$c_t(T^*Y \times_Y X) = c_t(T^*S \times_S X) c_t(f^*T^*(Y/S)). \tag{3.12.9}$$

It follows from (3.11.11) that

$$\overline{cc}_{X/Y}(\mathcal{F}) = (-1)^{r+s} \sum_{j\geq r} s_{j-r}(f^*T^*(Y/S)) \cap a_j. \tag{3.12.10}$$

The projection $T^*Y \times_Y X \longrightarrow f^*T^*(Y/S)$ has kernel $T^*S \times_S X$, and

$$(df)^{-1} SS(\mathcal{F}) \cap (T^*S \times_S X) \subset 0(X) \tag{3.12.11}$$

by the $SS(\mathcal{F})$-transversality of h, hence it is finite on $(df)^{-1}SS(\mathcal{F})$. Let

$$q_S : \mathbb{P}(f^*T^*(Y/S)) \longrightarrow X \tag{3.12.12}$$

be the projective bundle projection, and put $\xi = c_1(\mathcal{O}_{\mathbb{P}(f^*T^*(Y/S))}(1))$. The preceding finite projection induces a finite morphism

$$\rho : \mathbb{P}((df)^{-1} SS(\mathcal{F})) \longrightarrow \mathbb{P}(f^*T^*(Y/S)) \tag{3.12.13}$$

which identifies the tautological line bundles; hence we may slightly abuse the tautological bundle notation here. By the construction of the classes b_j we have

$$\rho_*(df^! CC(\mathcal{F})_{\mathcal{O}(1)}) = \sum_{j\geq 0} \xi^j \cap q_S^* a_j. \tag{3.12.14}$$

On $\mathbb{P}(f^*T^*(Y/S))$ we have the canonical quotient bundle $q_S^* f^*T^*(Y/S)/\mathcal{O}(-1)$, projective bundle formula gives

(3.12.15)

$$(q_S)_* \left[c_{r-1} \left(q_S^* f^*T^*(Y/S) - \mathcal{O}(-1) \right) \cap \xi^j \right] = \begin{cases} 1, & j = 0, \\ 0, & 0 < j < r, \\ -c_r(f^*T^*(Y/S))\, s_{j-r}(f^*T^*(Y/S)), & j \geq r. \end{cases}$$

Applying (3.12.15) and the definition of $cc^Z_{X/Y/S}(\mathcal{F})$, we obtain

$$\tau_* cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s \left[a_0 - c_r(f^*T^*(Y/S)) \cap \sum_{j\geq r} s_{j-r}(f^*T^*(Y/S)) \cap a_j \right]. \tag{3.12.16}$$

By (3.12.10), the second term on the right is $c_r\big((f^*T^*(Y/S))^\vee\big) \cap cc_{X/Y}(\mathcal{F})$. Together with $cc_{X/S}(\mathcal{F}) = (-1)^s a_0$, this proves (3.12.1).

It remains to prove the restriction assertion. Put $U = X \setminus Z$. On U, the morphism $f|_U : U \longrightarrow Y$ is $SS(\mathcal{F}|_U)$-transversal. Restricting (3.11.7) to U, the zero-section excess formula gives

$$\sum_j (b_j|_U)t^j = c_t(T^*Y \times_Y U)\sum_{j\geq 0} d_j t^j, \qquad d_0 = (df)^! CC(\mathcal{F}|_U). \tag{3.12.17}$$

Consequently,

$$cc_{X/Y}(\mathcal{F})|_U = (-1)^m d_0 = cc_{U/Y}(\mathcal{F}|_U). \tag{3.12.18}$$

This proves the restriction statement.

Finally, if $\dim Z < r+s$, then $\mathrm{CH}_{r+s}(Z) = 0$. Hence we have an isomorphism

$$\mathrm{CH}_{r+s}(X) \xrightarrow{\sim} \mathrm{CH}_{r+s}(X \setminus Z), \tag{3.12.19}$$

by the localization sequence, and $cc_{X/Y}(\mathcal{F})$ is the unique extension of $cc_{(X\setminus Z)/Y}(\mathcal{F}|_{X\setminus Z})$. Equation (3.12.3) is the special case $Z = \varnothing$. ☐

Remark 3.13. In fact, the construction of $\overline{cc}_{X/Y}(\mathcal{F})$ can be regarded as a generalized definition of the characteristic class, weakening the transversality condition.

Proposition 3.14. *Let $S = \mathrm{Spec}\, k$, $Y = \mathbb{A}^1_k$ with coordinate t, and let Z contain the characteristic locus of f. Writing $df : X \to T^*X$ for the section defined by dt, one has*

$$cc^Z_{X/\mathbb{A}^1_k/k}(\mathcal{F}) = (df)^! CC(\mathcal{F}) \quad \textit{in } \mathrm{CH}_0(Z). \tag{3.14.1}$$

Proof. The form dt identifies $\mathbb{P}(T^*\mathbb{A}^1_k \times_{\mathbb{A}^1_k} X)$ with X, trivializes $\mathcal{O}(-1)$, and identifies $\overline{df}$ with df. In (3.9.1), $r = 1$, $s = 0$ and the Chern class involved is $c_0 = 1$. ☐

Theorem 3.15 (Saito's isolated Milnor formula)**.** *Assume $S = \mathrm{Spec}\, k$ and Y is a smooth curve. Let Z be a finite set of closed points of X, and assume f is $SS(\mathcal{F})$-transversal outside Z. Then*

$$cc^Z_{X/Y/k}(\mathcal{F}) = -\sum_{x\in Z} \mathrm{dimtot}\, R\Phi_{\bar{x}}(\mathcal{F}, f)\,[x] \quad \textit{in } \mathrm{CH}_0(Z). \tag{3.15.1}$$

Here total dimension is computed after derived reduction of Λ to its residue field, with alternating signs for complexes.

Proof. Near $f(x)$ choose an étale coordinate on Y, we may assume $Y = \mathbb{A}^1_k$. The consequence follows from Saito's Milnor formula [14, Theorem 5.9]. ☐

We expect the following Milnor type formula for non-isolated singular/characteristic points holds.

Conjecture 3.16. *Let S be a smooth connected k-scheme of dimension s. Consider the commutative diagram (3.8.1). Let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$ such that $X \setminus Z \to Y$ is $SS(\mathcal{F}|_{X\setminus Z})$-transversal and that $X \to S$ is $SS(\mathcal{F})$-transversal. Then we have an equality*

$$\widetilde{C}^Z_{X/Y/S}(\mathcal{F}) = \widetilde{\mathrm{cl}}(cc^Z_{X/Y/S}(\mathcal{F})) \qquad \text{in} \qquad H^0_Z(X, \mathcal{K}_{X/Y/S}), \tag{3.16.1}$$

where $\widetilde{\mathrm{cl}}$ is the composition $\mathrm{CH}_s(Z) \xrightarrow{\mathrm{cl}} H^0_Z(X, \mathcal{K}_{X/S}) \xrightarrow{(3.2.3)} H^0_Z(X, \mathcal{K}_{X/Y/S})$.

When $S = \mathrm{Spec}\, k$, $Y = \mathbb{A}^1_k$ and $Z = \{x\}$, then Conjecture 3.16 follows from Saito's isolated Milnor formula (3.15.1) and the cohomological Milnor formula (3.3.4).

Remark 3.17. Let $f : X \to Y$ be a separated morphism between smooth schemes over k. Let $Z \subseteq X$ be a closed subset and $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$. Assume that f is universally locally acyclic relatively to $\mathcal{F}$ outside Z. Let $n = \dim X$. We expect that there is a n-cycle $CC^Z_{X/Y}(\mathcal{F})$ supported on $T^*X \times_X Z$ such that

$$\widetilde{\mathrm{cl}}(0^!_X CC^Z_{X/Y}(\mathcal{F})) = \widetilde{C}^Z_{X/Y/k}(\mathcal{F}) \quad \text{in} \quad H^0_Z(X, \mathcal{K}_{X/Y/k}). \tag{3.17.1}$$

If Y is a smooth curve and Z is a finite set of closed points of X, then

$$CC^Z_{X/Y}(\mathcal{F}) = -\sum_{x\in Z} \operatorname{dimtot} R\Phi_{\bar{x}}(\mathcal{F}, f)\cdot [T^*_x X]. \tag{3.17.2}$$

If $f = \mathrm{id}$ and Z is the smallest closed subset of X such that $\mathcal{F}|_{X\setminus Z}$ is smooth, then

$$CC^Z_{X/Y}(\mathcal{F}) = CC(\mathcal{F}) - \mathrm{rank}\mathcal{F}\cdot CC(\Lambda). \tag{3.17.3}$$

In order to construct $CC^Z_{X/Y}(\mathcal{F})$, one shall introduce a singular support with respect to a morphism $f : X \to Y$. When f is the identity morphism, then id-singular support is the singular support defined by Beilinson [4]. We expect the f-singular support to exist under suitable conditions and the non-acyclicity cycle $CC^Z_{X/Y}(\mathcal{F})$ to be supported on this f-relative singular support.

Proposition 3.18. *Consider a Cartesian diagram in* Sm_k

$$\begin{tikzcd} X' \arrow[r, "i"] \arrow[d, "f'"'] \arrow[dr, phantom, "\square"] & X \arrow[d, "f"] \\ S' \arrow[r, "\delta"'] & S. \end{tikzcd} \tag{3.18.1}$$

We assume that f and f' are smooth morphisms, and S and S' are connected of dimension s and s' respectively. Let $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$. Assume that $X\to S$ is $SS(\mathcal{F})$-transversal and i is properly $SS(\mathcal{F})$-transversal. Then we have

$$i^! cc_{X/S}(\mathcal{F}) = cc_{X'/S'}(i^*\mathcal{F}) \quad \text{in} \quad \mathrm{CH}_{s'}(X'), \tag{3.18.2}$$

where $i^!\colon \mathrm{CH}_s(X)\to \mathrm{CH}_{s'}(X')$ is the refined Gysin pullback.

Proof. We consider the following diagram
(3.18.3)

$$\begin{tikzcd}
{} & T^*X' \arrow[dd] & & T^*X\times_X X' \arrow[ll, "di"'] \arrow[rr, "\mathrm{pr}"] \arrow[dd] & & T^*X \arrow[dd] \\
T^*S'\times_{S'} X' \arrow[ur] \arrow[dd] & & T^*S\times_S X' \arrow[ll, "d\delta"'] \arrow[ur] \arrow[rr] \arrow[dd] & & T^*S\times_S X \arrow[ur] \arrow[dd] & \\
& T^*(X'/S') \arrow[rr, "\simeq"] & & T^*(X/S)\times_X X' \arrow[rr] & & T^*(X/S) \\
X' \arrow[ur, "0_{X'/S'}"] \arrow[rr, "="] & & X' \arrow[ur, "0_{X/S}\times 1"] \arrow[rr, "i"] & & X \arrow[ur, "0_{X/S}"'] &
\end{tikzcd}$$

Note that the square containing the morphisms di and $d\delta$ is Cartesian. In the following calculations, although di and $d\delta$ are not proper, we can still apply di_* and $d\delta_*$ since di is finite on the support of $\mathrm{pr}^{-1}(SS(\mathcal{F}))$ and $d\delta$ is finite on the zero section X' of $T^*S\times_S X'$. We have

$$\begin{aligned} cc_{X'/S'}(i^*\mathcal{F}) &= (-1)^{s'}\cdot 0^!_{X'/S'}CC(i^*\mathcal{F}) \\ &\overset{(a)}{=} (-1)^{s'}\cdot 0^!_{X'/S'}(di_*\mathrm{pr}^!CC(\mathcal{F})\cdot(-1)^{-\dim(X')+\dim(X)}) \\ &= d\delta_* 0^!_{X'/S'}\mathrm{pr}^!CC(\mathcal{F})\cdot(-1)^{s'-\dim(X')+\dim(X)} \\ &\overset{(b)}{=} 0^!_{X'/S'}\mathrm{pr}^!CC(\mathcal{F})\cdot(-1)^{s} = 0^!_{X/S}\mathrm{pr}^!CC(\mathcal{F})\cdot(-1)^{s} \\ &= i^!(0^!_{X/S}CC(\mathcal{F})\cdot(-1)^s) = i^! cc_{X/S}(\mathcal{F}). \end{aligned} \tag{3.18.4}$$

where (a) follows from [14, Theorem 7.6] and (b) follows from the fact that $0^!_{X'/S'}\mathrm{pr}^!CC(\mathcal{F})$ is supported on the zero section of $T^*S\times_S X'$. □

3.19. Consider a commutative diagram in Sm_k:

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "h"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd} \tag{3.19.1}$$

Let $\mathcal{F}\in D_{\mathrm{ctf}}(X,\Lambda)$ such that h is $SS(\mathcal{F})$-transversal. Assume f is proper on $\mathrm{supp}(\mathcal{F})$. By [14, Lemma 3.8], g is $f_\circ SS(\mathcal{F})$-transversal. By [4, Lemma 2.2(ii)], $SS(Rf_*\mathcal{F})\subseteq f_\circ SS(\mathcal{F})$. Thus $g: Y\to S$ is also $SS(Rf_*\mathcal{F})$-transversal and the class $cc_{Y/S}(Rf_*\mathcal{F})$ is well-defined.

Proposition 3.20. *Consider the assumptions in §3.19. Assume moreover that Y is projective, $f: X\to Y$ is quasi-projective and $\dim f_\circ SS(\mathcal{F})\le \dim Y$. Then we have*

$$f_*cc_{X/S}(\mathcal{F}) = cc_{Y/S}(Rf_*\mathcal{F}) \quad \text{in} \quad \mathrm{CH}_s(Y). \tag{3.20.1}$$

We don't know how to remove the assumption $\dim f_\circ SS(\mathcal{F})\le \dim Y$.

Proof. Consider the following commutative diagram

$$\begin{tikzcd} & X \arrow[r, "f"] \arrow[d, hook, "0_X"] & Y \arrow[d, hook, "0_Y"] \\ & T^*S\times_S X \arrow[r, "\mathrm{id}\times f"] \arrow[dl, "dh"'] \arrow[d, "r"] & T^*S\times_S Y \arrow[d, "dg"] \\ T^*X & T^*Y\times_Y X \arrow[l, "df"] \arrow[r, "\mathrm{id}\times f"] & T^*Y \end{tikzcd} \tag{3.20.2}$$

Then we have

$$\begin{aligned} f_*cc_{X/S}(\mathcal{F}) &\overset{(3.5.2)}{=} (-1)^s\cdot f_*dh^!(CC(\mathcal{F})) = (-1)^s\cdot(\mathrm{id}\times f)_*r^!df^!(CC(\mathcal{F})) \\ &= (-1)^s\cdot dg^!(\mathrm{id}\times f)_*df^!(CC(\mathcal{F})) \\ &\overset{(a)}{=} (-1)^s\cdot dg^!CC(Rf_*\mathcal{F}) \overset{(3.5.2)}{=} cc_{Y/S}(Rf_*\mathcal{F}). \end{aligned} \tag{3.20.3}$$

where (a) follows from [15, Theorem 2.2.5]. □

Lemma 3.21. *On an open neighbourhood of $\mathrm{supp}(\mathcal{F})$, the morphism $h: X\to S$ is smooth. Let*

$$\pi_{X/S}: T^*X \longrightarrow T^*(X/S), \qquad q_S: \mathbb{P}(f^*T^*(Y/S))\longrightarrow X \tag{3.21.1}$$

be the natural projections on this neighbourhood. The restriction of $\pi_{X/S}$ to $SS(\mathcal{F})$ is finite. The relative differential $\mathcal{O}(-1)\to q_S^*T^*(X/S)$ defines a section

$$\overline{d_{/S}f}: \mathbb{P}(f^*T^*(Y/S))\longrightarrow q_S^*T^*(X/S)\otimes\mathcal{O}(1), \tag{3.21.2}$$

and we have

$$cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s(q_S)_*\left[c_{r-1}\left(\frac{q_S^*f^*T^*(Y/S)}{\mathcal{O}(-1)}\right)\cap\overline{d_{/S}f}^{\,!}\left((\pi_{X/S})_*CC(\mathcal{F})\right)_{\mathcal{O}(1)}\right]. \tag{3.21.3}$$

Here $(\pi_{X/S})_*$ is taken on the characteristic-cycle support, and $(q_S)_*$ is taken on the reduced intersection support over Z.

Proof. For $x \in \operatorname{supp}(\mathcal{F})$, we have $0_x \in SS(\mathcal{F})$. The $SS(\mathcal{F})$-transversality of h therefore implies that $dh_x : T^*_{h(x)}S \longrightarrow T^*_x X$ is injective. Hence h is smooth on an open neighbourhood of $\operatorname{supp}(\mathcal{F})$, and we have

$$0 \longrightarrow T^*S \times_S X \xrightarrow{dh} T^*X \xrightarrow{\pi_{X/S}} T^*(X/S) \longrightarrow 0. \tag{3.21.4}$$

on the open neighbourhood. Moreover,

$$SS(\mathcal{F}) \cap \ker(\pi_{X/S}) \subset 0(X), \tag{3.21.5}$$

so $\pi_{X/S}|_{SS(\mathcal{F})}$ is finite by [4, proof of Lemma 1.2(ii)].

The following square is Cartesian, since its vertical kernels are identified by dh:

$$\begin{tikzcd} T^*Y \times_Y X \arrow[r, "df"] \arrow[d] & T^*X \arrow[d, "\pi_{X/S}"] \\ f^*T^*(Y/S) \arrow[r, "d_{/S}f"'] & T^*(X/S). \end{tikzcd} \tag{3.21.6}$$

The tautological line bundle therefore gives

$$\mathcal{O}(-1) \longrightarrow q_S^* f^* T^*(Y/S) \longrightarrow q_S^* T^*(X/S), \tag{3.21.7}$$

hence we have a section $\overline{d_{/S}f}$ as in the statement. Since $\pi_{X/S}$ is finite on $SS(\mathcal{F})$, Gysin pullback commutes with proper pushforward [10, Theorem 6.2], hence we have

$$\rho_* \, df^! CC(\mathcal{F}) = (d_{/S}f)^! (\pi_{X/S})_* CC(\mathcal{F}). \tag{3.21.8}$$

By transversality,

$$(df)^{-1} SS(\mathcal{F}) \cap (T^*S \times_S X) \subset 0(X), \tag{3.21.9}$$

hence $T^*Y \times_Y X \longrightarrow f^*T^*(Y/S)$ induces a finite map of projective intersection supports

$$\bar{\rho} : \mathbb{P}((df)^{-1} SS(\mathcal{F})) \longrightarrow \mathbb{P}((d_{/S}f)^{-1}(\pi_{X/S} SS(\mathcal{F}))), \tag{3.21.10}$$

and $\bar{\rho}^* \mathcal{O}(-1) = \mathcal{O}(-1)$. Now by calculating on open covers trivializing $\mathcal{O}(-1)$, we have

$$\bar{\rho}_* (\overline{df}^! CC(\mathcal{F})_{\mathcal{O}(1)}) = \overline{d_{/S}f}^! ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)}. \tag{3.21.11}$$

Hence

$$\bar{\rho}^* \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) = \frac{q^* f^* T^*(Y/S)}{\mathcal{O}(-1)}, \tag{3.21.12}$$

and the projection formula and (3.9.1) therefore implies

$$cc^Z_{X/Y/S}(\mathcal{F}) = (-1)^s (q_S)_* \left[c_{r-1} \left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)} \right) \cap \overline{d_{/S}f}^! ((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)} \right]. \tag{3.21.13}$$

which is (3.21.3). □

Proposition 3.22. *Consider a commutative diagram in* Sm_k

$$\begin{tikzcd} X' \arrow[r, "i_X"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "i_Y"] \arrow[d, "g'"'] & Y \arrow[d, "g"] \\ S' \arrow[r, "\delta"'] & S, \end{tikzcd} \tag{3.22.1}$$

whose squares are Cartesian. Put $Z' = Z \times_X X'$, and write $s' = \dim S'$. Assume that X', Y', S' are smooth and pure-dimensional, that S' is connected, that $\dim X' - \dim X = s' - s$, and that i_X is properly $SS(\mathcal{F})$-transversal. Then

$$i_X^! cc^Z_{X/Y/S}(\mathcal{F}) = cc^{Z'}_{X'/Y'/S'}(i_X^*\mathcal{F}) \quad \text{in } \mathrm{CH}_{s'}(Z'). \tag{3.22.2}$$

Proof. Using the Cartesian diagram (3.22.1), proper $SS(\mathcal{F})$-transversality gives

$$SS(i_X^*\mathcal{F}) \subset di_X(\mathrm{pr}^{-1}SS(\mathcal{F})), \qquad \mathrm{pr} : T^*X \times_X X' \longrightarrow T^*X. \tag{3.22.3}$$

Hence

$$SS(i_X^*\mathcal{F}) \cap (T^*S' \times_{S'} X') \subset 0(X'), \qquad (df')^{-1}SS(i_X^*\mathcal{F}) \subset 0(X') \cup (T^*Y' \times_{Y'} Z'). \tag{3.22.4}$$

Thus $g' \circ f'$ is $SS(i_X^*\mathcal{F})$-transversal and $f'|_{X' \setminus Z'}$ is $SS((i_X^*\mathcal{F})|_{X'\setminus Z'})$-transversal. Pullback formula of characteristic cycles [14, Theorem 7.6] gives

$$CC(i_X^*\mathcal{F}) = (-1)^{\dim X' - \dim X}(di_X)_*\mathrm{pr}^! CC(\mathcal{F}), \tag{3.22.5}$$

then by base change we have

$$(\pi_{X'/S'})_* CC(i_X^*\mathcal{F}) = (-1)^{\dim X' - \dim X}\, i_{X/S}^!(\pi_{X/S})_* CC(\mathcal{F}), \tag{3.22.6}$$

where

$$i_{X/S} : T^*(X'/S') \simeq T^*(X/S) \times_X X' \longrightarrow T^*(X/S) \tag{3.22.7}$$

is the natural projection. Let $i_{\mathbb{P}} : \mathbb{P}(f'^*T^*(Y'/S')) \longrightarrow \mathbb{P}(f^*T^*(Y/S))$ be the induced base change. Then

$$\begin{tikzcd} \mathbb{P}(f'^*\Omega^1_{Y'/S'}) \arrow[r, "i_{\mathbb{P}}"] \arrow[d, "q_{S'}"'] & \mathbb{P}(f^*\Omega^1_{Y/S}) \arrow[d, "q_S"] \\ X' \arrow[r, "i_X"'] & X \end{tikzcd} \tag{3.22.8}$$

is Cartesian, and $i_{\mathbb{P}}^*\mathcal{O}(-1) = \mathcal{O}(-1)$, $i_{\mathbb{P}}^* q_S^* f^* T^*(Y/S) = q_{S'}^* f'^* T^*(Y'/S')$. Thus

$$i_{\mathbb{P}}^*\left(\frac{q_S^* f^* T^*(Y/S)}{\mathcal{O}(-1)}\right) = \frac{q_{S'}^* f'^* T^*(Y'/S')}{\mathcal{O}(-1)}. \tag{3.22.9}$$

The relative differentials also commute with base change:

$$\begin{tikzcd} \mathbb{P}(f'^*T^*(Y'/S')) \arrow[r, "i_{\mathbb{P}}"] \arrow[d, "\overline{d_{/S'}f'}"'] & \mathbb{P}(f^*T^*(Y/S)) \arrow[d, "\overline{d_{/S}f}"] \\ q_{S'}^*T^*(X'/S') \otimes \mathcal{O}(1) \arrow[r, "\sim"'] & i_{\mathbb{P}}^*(q_S^*T^*(X/S) \otimes \mathcal{O}(1)). \end{tikzcd} \tag{3.22.10}$$

Hence base change of Gysin pullback gives

$$\overline{d_{/S'}f'}^{\,!}((\pi_{X'/S'})_* CC(i_X^*\mathcal{F}))_{\mathcal{O}(1)} = (-1)^{\dim X' - \dim X} i_{\mathbb{P}}^! \left[\overline{d_{/S}f}^{\,!}((\pi_{X/S})_* CC(\mathcal{F}))_{\mathcal{O}(1)}\right]. \tag{3.22.11}$$

Applying (3.21.3) and $i_X^!(q_S)_* = (q_{S'})_* i_{\mathbb{P}}^!$ we have

$$cc^{Z'}_{X'/Y'/S'}(i_X^*\mathcal{F}) = (-1)^{s' + \dim X' - \dim X - s}\, i_X^! cc^Z_{X/Y/S}(\mathcal{F}). \tag{3.22.12}$$

Since $\dim X' - \dim X = s' - s$, we have (3.22.2). □

In particular, if i_X is flat of pure relative dimension $s'-s$, then

$$\mathrm{pr}^{-1}SS(\mathcal{F})\cap\ker(di_X)\subset 0(X'), \tag{3.22.13}$$

so i_X is $SS(\mathcal{F})$-transversal; moreover, for every irreducible component $C\subset SS(\mathcal{F})$,

$$\dim(C\times_X X')=\dim C+s'-s=\dim X+s'-s=\dim X', \tag{3.22.14}$$

hence i_X is properly $SS(\mathcal{F})$-transversal.

Proposition 3.23. *Let $p: X\to X'$ be a proper morphism in Sm_k, with X' of pure dimension, and let $f': X'\to Y$ satisfy $f=f'\circ p$. Let $Z'\subseteq X'$ be closed, with $Z\subseteq p^{-1}(Z')$. Assume that $\dim p_\circ SS(\mathcal{F})\leq\dim X'$, then $g\circ f'$ is $SS(Rp_*\mathcal{F})$-transversal, and f' is $SS(Rp_*\mathcal{F})$-transversal outside Z'. Moreover,*

$$p_*cc^{Z}_{X/Y/S}(\mathcal{F})=cc^{Z'}_{X'/Y/S}(Rp_*\mathcal{F})\quad\textit{in }\mathrm{CH}_s(Z'). \tag{3.23.1}$$

Proof. Proper pushforward of singular support gives $SS(Rp_*\mathcal{F})\subseteq p_\circ SS(\mathcal{F})$ [4, Lemma 2.2(ii)]. A covector in $p_\circ SS(\mathcal{F})$ lifts to a covector whose pullback by dp belongs to $SS(\mathcal{F})$. The hypotheses on h and f therefore give

$$\begin{aligned}(df')^{-1}p_\circ SS(\mathcal{F})&\subseteq 0(X')\cup(T^*Y\times_Y Z'),\\ (df')^{-1}p_\circ SS(\mathcal{F})\cap(T^*S\times_S X')&\subseteq 0(X').\end{aligned} \tag{3.23.2}$$

Here $0(X')$ is the zero section of $T^*Y\times_Y X'$. This proves both transversality assertions, and also gives a common projective support on which the quotient bundle is defined.

Abe's proper-pushforward theorem [2, Introduction, Theorem; §6.6] gives

$$CC(Rp_*\mathcal{F})=p_!CC(\mathcal{F})=\mathrm{pr}_{2*}(dp)^!CC(\mathcal{F}), \tag{3.23.3}$$

initially after inverting $\mathrm{char}\,k$. Let q' be the projective projection over X', and let

$$p_{\mathbb{P}}:\mathbb{P}(T^*Y\times_Y X)\longrightarrow\mathbb{P}(T^*Y\times_Y X') \tag{3.23.4}$$

be the base change of p. We have natural maps

$$\begin{aligned}dp\otimes 1: q^*(T^*X'\times_{X'}X)\otimes\mathcal{O}(1)&\longrightarrow q^*T^*X\otimes\mathcal{O}(1),\\ \rho: q^*(T^*X'\times_{X'}X)\otimes\mathcal{O}(1)&\longrightarrow q'^*T^*X'\otimes\mathcal{O}(1).\end{aligned} \tag{3.23.5}$$

The second map is proper. Let

$$D\subset q'^*T^*X'\otimes\mathcal{O}(1) \tag{3.23.6}$$

be the $\mathcal{O}(1)$-twist of the pullback of $p_\circ SS(\mathcal{F})$, then $\dim D\leq\dim X'+\dim Y-1$. On an open subset trivializing $\mathcal{O}(-1)$ we have

$$\rho_*(dp\otimes 1)^!CC(\mathcal{F})_{\mathcal{O}(1)}=CC(Rp_*\mathcal{F})_{\mathcal{O}(1)}\qquad\text{in }\mathrm{CH}_{\dim X'+\dim Y-1}(D)\left[\frac{1}{\mathrm{char}\,k}\right]. \tag{3.23.7}$$

Since $\mathrm{CH}_{\dim X'+\dim Y-1}(D)=Z_{\dim X'+\dim Y-1}(D)$ and inverting $\mathrm{char}\,k$ is injective on the free abelian group $Z_{\dim X'+\dim Y-1}(D)$, the local equality (3.23.7) is an equality of cycles and hence glue globally into an equality in $Z_{\dim X'+\dim Y-1}(D)$. Let $p_{\mathbb{P}}$ be the base change of p. The pullback of df' is a section

$$p_{\mathbb{P}}^*df':\mathbb{P}(T^*Y\times_Y X)\longrightarrow q^*(T^*X'\times_{X'}X)\otimes\mathcal{O}(1), \tag{3.23.8}$$

and

$$(dp\otimes 1)\circ p_{\mathbb{P}}^*df'=df,\qquad\rho\circ p_{\mathbb{P}}^*df'=df'\circ p_{\mathbb{P}}. \tag{3.23.9}$$

Therefore, by compatibility of refined Gysin pullback with proper pushforward [10, Theorem 6.2],

$$\begin{aligned}(p_{\mathbb{P}})_* df^! CC(\mathcal{F})_{\mathcal{O}(1)} &= (p_{\mathbb{P}})_*(p_{\mathbb{P}}^* df')^!(dp\otimes 1)^! CC(\mathcal{F})_{\mathcal{O}(1)} \\ &= df'^!\, \rho_*(dp\otimes 1)^! CC(\mathcal{F})_{\mathcal{O}(1)} \\ &= df'^! CC(Rp_*\mathcal{F})_{\mathcal{O}(1)}.\end{aligned} \tag{3.23.10}$$

Combining with

$$p_{\mathbb{P}}^*\left(\frac{q'^* f'^* T^*(Y/S)}{\mathcal{O}(-1)}\right) = \frac{q^* f^* T^*(Y/S)}{\mathcal{O}(-1)}, \tag{3.23.11}$$

we have

$$\begin{aligned}p_* cc^Z_{X/Y/S}(\mathcal{F}) &= (-1)^s q'_*\left[c_{r-1}\left(\frac{q'^* f'^* \Omega^1_{Y/S}}{\mathcal{O}(-1)}\right)\cap df'^! CC(Rp_*\mathcal{F})_{\mathcal{O}(1)}\right] \\ &= cc^{Z'}_{X'/Y/S}(Rp_*\mathcal{F}),\end{aligned} \tag{3.23.12}$$

which proves the assertion. □

Corollary 3.24 (Conductor formula). *Let $S = \mathrm{Spec}(k)$, let Y be a smooth connected curve, and let $f : X \to Y$ be proper. Let $T \subset Y$ be a finite set of closed points, and put $Z = f^{-1}(T)$. Assume that f is $SS(\mathcal{F})$-transversal outside Z. Then*

$$f_* cc^Z_{X/Y/k}(\mathcal{F}) = -\sum_{y\in T} a_y(Rf_*\mathcal{F})[y] \quad \text{in } \mathrm{CH}_0(T), \tag{3.24.1}$$

where a_y is the Artin conductor, computed after derived reduction of the coefficient ring to its residue field.

Proof. We may assume $Y = \mathbb{A}^1_k$ and $T = \{0\}$. Combining Proposition 3.14, Theorem 3.15 and Proposition 3.23 we get the desired result. □

4. non-isolated Milnor formula in quasi-affine case

4.1. Let k be an algebraically closed field of characteristic $p > 0$, Λ be a finite local ring such that the characteristic of the residue field is invertible in k. Let X be a smooth separated scheme of finite type over k purely of dimension d, let $f : X \to \mathbb{A}^1_k$ be a k-morphism and let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$.

Let t be the standard coordinate on $\mathbb{A}^1_k$. The form dt defines

$$df : X \to T^*X \tag{4.1.1}$$

which is a section of T^*X. Put $Z = Z_{X,f} := (df)^{-1}SS(\mathcal{F})$, [14, Lemma 4.2(5)] implies f is universally locally acyclic relatively to $\mathcal{F}$ in $X \setminus Z$.

Theorem 4.2. *Consider the previous assumptions. Assume that X is quasi-affine. Then we have*

$$\widetilde{C}^Z_{X/\mathbb{A}^1_k/k}(\mathcal{F}) = \widetilde{\mathrm{cl}}\left(df^! CC(\mathcal{F})\right). \tag{4.2.1}$$

By Proposition 3.14, $cc^Z_{X/\mathbb{A}^1_k/k}(\mathcal{F}) = df^! CC(\mathcal{F})$ in $\mathrm{CH}_0(Z)$. Hence Theorem 4.2 verifies Conjecture 3.16 in the quasi-affine case.

We devote the rest of this section to the proof of the theorem.

4.3. We first recall Fulton's construction of specialization over a discrete valuation ring [10, Chapter 20]. Let R be a discrete valuation ring, let $S = \operatorname{Spec} R$ with $j : \eta \hookrightarrow S$, $i : s \hookrightarrow S$ be its generic point and closed point, respectively. Let u be a uniformizer of R.

Define the dimension function by setting $\delta_S(s) = 0$, $\delta_S(\eta) = 1$ and

$$\delta_T(x) = \delta_S(p(x)) + \operatorname{trdeg}_{\kappa(p(x))} \kappa(x) \tag{4.3.1}$$

for a morphism $p : T \to S$ locally of finite type with $x \in T$. For an integral closed subscheme $V \subset T$, its S-dimension is defined to be $\delta_T(\xi_V)$ where ξ_V is the generic point of V. Let $Z_m(T/S)$ be the free abelian group on integral closed subschemes of S-dimension m, we may define rational equivalence in the usual way: it is generated by $\operatorname{div}_V(h)$ for rational functions h on integral closed V of S-dimension $m+1$. Put $\mathrm{CH}_m(T/S) = Z_m(T/S)/\sim_{\mathrm{rat}}$. Let $j_T : T_\eta \to T$, $i_T : T_s \to T$ be the base changes of j and i, respectively. Then, with a slight abuse of notation, we have

$$j_T^* : \mathrm{CH}_{m+1}(T/S) \longrightarrow \mathrm{CH}_m(T_\eta), \qquad i_T^! : \mathrm{CH}_{m+1}(T/S) \longrightarrow \mathrm{CH}_m(T_s). \tag{4.3.2}$$

The second map is the refined Gysin map obtained by the regular immersion i.

Moreover, we have the localization exact sequence

$$\mathrm{CH}_{r+1}(T_s) \xrightarrow{i_{T*}} \mathrm{CH}_{r+1}(T/S) \xrightarrow{j_T^*} \mathrm{CH}_r(T_\eta) \longrightarrow 0. \tag{4.3.3}$$

For any $\alpha \in \mathrm{CH}_r(T_\eta)$, choose a lift $\widetilde{\alpha} \in \mathrm{CH}_{r+1}(T/S)$ by (4.3.3) and define

$$\mathrm{sp}^{\mathrm{cyc}}(\alpha) = i_T^!(\widetilde{\alpha}) \in \mathrm{CH}_r(T_s). \tag{4.3.4}$$

One may check $\mathrm{sp}^{\mathrm{cyc}}(\alpha)$ is independent of the lift. Indeed, two lifts differ by some $i_{T*}\xi$, and the refined self-intersection formula gives

$$i_T^! i_{T*}\xi = c_1(p_s^* N_{s/S}) \cap \xi = 0, \tag{4.3.5}$$

because $N_{s/S} \simeq \mathcal{O}_s$. In particular,

$$\mathrm{sp}^{\mathrm{cyc}}(j_T^*\gamma) = i_T^!\gamma \tag{4.3.6}$$

for $\gamma \in \mathrm{CH}_{r+1}(T/S)$. For integral subscheme $V \subset T$ such that $V_\eta \neq \varnothing$ (in this case we call V *horizontal*), $\mathrm{sp}^{\mathrm{cyc}}([V_\eta])$ is the divisor $\operatorname{div}_V(u)$, with the usual length multiplicities along the special fiber.

4.4. Let B be a smooth scheme separated of finite type over S. Let $E \to B$ be a vector bundle of rank r, let $\sigma : B \to E$ be a section, let C be a closed subscheme of E, and put $Z = B \times_E C$. Since σ is a regular immersion of codimension r, there is a supported refined Gysin map

$$\sigma^! : \mathrm{CH}_{m+1}(C/S) \longrightarrow \mathrm{CH}_{m-r+1}(Z/S). \tag{4.4.1}$$

Lemma 4.5. *Consider the assumptions in §4.4. Let $\gamma \in \mathrm{CH}_{m+1}(C/S)$. Define*

$$\gamma_\eta = j_C^*\gamma \in \mathrm{CH}_m(C_\eta), \qquad \gamma_s = i_C^!\gamma \in \mathrm{CH}_m(C_s). \tag{4.5.1}$$

Then we have

$$j_Z^*(\sigma^!\gamma) = \sigma_\eta^!\gamma_\eta \quad \textit{in } \mathrm{CH}_{m-r}(Z_\eta), \tag{4.5.2}$$

$$\mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^!\gamma_\eta) = \sigma_s^!\gamma_s \quad \textit{in } \mathrm{CH}_{m-r}(Z_s). \tag{4.5.3}$$

Proof. Refined Gysin pullback exchanges with flat base change $\eta \to S$, hence we have

$$j_Z^* \circ \sigma^! = \sigma_\eta^! \circ j_C^*, \tag{4.5.4}$$

which proves (4.5.2).

For the special fiber, we have

$$\mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^!\gamma_\eta) \overset{(4.5.2)}{=} \mathrm{sp}^{\mathrm{cyc}}(j_Z^*\sigma^!\gamma) \overset{(4.3.6)}{=} i_Z^!\sigma^!\gamma \overset{[10,\,\S6.4]}{=} \sigma_s^! i_C^!\gamma. \tag{4.5.5}$$

□

Corollary 4.6. *Consider the assumptions in Lemma 4.5. Suppose*

$$\alpha = \sum_a n_a[C_a] \tag{4.6.1}$$

is a cycle supported on C, with components $C_a \subset E$ flat over S of pure relative dimension m, and put $q = m - r \geq 0$. For $t \in \{\eta, s\}$ define

$$\alpha_t := \sum_{a:(C_a)_t \neq \varnothing} n_a[(C_a)_t], \tag{4.6.2}$$

with the usual scheme-theoretic multiplicities at its generic points. Then

$$\mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^!\alpha_\eta) = \sigma_s^!\alpha_s \qquad \text{in } \mathrm{CH}_q(Z_s). \tag{4.6.3}$$

Proof. Since each C_a is flat over S with pure m-dimension fibers, we have a natural lift $\widetilde{\alpha} = \sum_a n_a[C_a]$ in $\mathrm{CH}_{m+1}(C/S)$. Flat pullback gives $j_C^*\widetilde{\alpha} = \alpha_\eta$, while Gysin pullback along $s \to S$ gives $i_C^!\widetilde{\alpha} = \alpha_s$. The consequence follows from Lemma 4.5. □

4.7. In §§4.7–4.12, let $S = \operatorname{Spec} R$ be a strictly henselian trait with generic point η and closed point s, and let $u \in R$ be a uniformizer. Let $f : X \to S$ be a smooth separated morphism of finite type and pure relative dimension d. Let $k : Z \to X$ be a closed subscheme. Let $j_X : X_\eta \to X$, $i_X : X_s \to X$ be the generic and special fiber, respectively, and $b_X : X_{\overline{\eta}} \to X_\eta$ be the base change to the algebraic closure $\overline{\eta}$. We use the convention of nearby cycle functor in [19, §3.4]:

$$R\Psi_f = i_X^* Rj_{X*} b_{X*} b_X{}^* : D^+(X_\eta, \Lambda) \to D^+(X_s, \Lambda). \tag{4.7.1}$$

For $0 \leq r \leq d$, $c = d - r$ and $t \in \{\eta, s\}$, we have the refined cycle class map

$$\mathrm{cl}_t : \mathrm{CH}_r(Z_t) \longrightarrow H^{2c}_{Z_t}(X_t, \Lambda(c)) \simeq H^{-2r}_{Z_t}(X_t, \mathcal{K}_{X_t/t}(-r)) \tag{4.7.2}$$

by Riou [13, Exposé XVI, Definition 2.3.1, Proposition 2.3.7, Definition 2.4.1].

Using the method of [19, §3.5], we have a natural specialization map on étale cohomology groups

$$\mathrm{sp}^{\text{ét}} : H^{-2r}_{Z_\eta}(X_\eta, \mathcal{K}_{X_\eta/\eta}(-r)) \longrightarrow H^{-2r}_{Z_s}(X_s, \mathcal{K}_{X_s/s}(-r)). \tag{4.7.3}$$

Theorem 4.8. *For every $0 \leq r \leq d$ and $c = d - r$, the square*

$$\begin{array}{ccc} \mathrm{CH}_r(Z_\eta) & \xrightarrow{\mathrm{cl}_\eta} & H^{2c}_{Z_\eta}(X_\eta, \Lambda(c)) \\ {\scriptstyle \mathrm{sp}^{\mathrm{cyc}}}\downarrow & & \downarrow{\scriptstyle \mathrm{sp}^{\text{ét}}} \\ \mathrm{CH}_r(Z_s) & \xrightarrow{\mathrm{cl}_s} & H^{2c}_{Z_s}(X_s, \Lambda(c)) \end{array} \tag{4.8.1}$$

commutes.

We postpone the proof of Theorem 4.8 to §4.11.

Lemma 4.9. *Let $W \subset Z$ be closed and $\gamma \in H^{2a}_W(X, \mathcal{K}_{X/S}(a))$, we have*

$$\mathrm{sp}^{\text{ét}}(j_X^*\gamma) = i_X^*\gamma \quad \text{in} \quad H^{2a}_{W_s}(X_s, \mathcal{K}_{X_s/s}(a)). \tag{4.9.1}$$

Proof. Let $\ell : W \to X$ be the closed immersion. By definition, γ is equivalent to a morphism $\gamma : \ell_*\Lambda_W \to \mathcal{K}_{X/S}(a)[2a]$ and we have

$$\ell_{s*}\Lambda_{W_s} \simeq i_X^*\ell_*\Lambda_W \longrightarrow R\Psi_f j_X^*\ell_*\Lambda_W \simeq R\Psi_f(\ell_{\eta*}\Lambda_{W_\eta}), \tag{4.9.2}$$

$$i_X^*\mathcal{K}_{X/S}(a)[2a] \longrightarrow R\Psi_f(j_X^*\mathcal{K}_{X/S})(a)[2a] \simeq R\Psi_f(\mathcal{K}_{X_\eta/\eta})(a)[2a]. \tag{4.9.3}$$

by proper base change. Moreover, we have the following commutative square

$$\begin{tikzcd} \ell_{s*}\Lambda_{W_s} \arrow[r, "(4.9.2)"] \arrow[d, "i_X^*\gamma"'] & R\Psi_f(\ell_{\eta*}\Lambda_{W_\eta}) \arrow[d, "R\Psi_f(j_X^*\gamma)"] \\ i_X^*\mathcal{K}_{X/S}(a)[2a] \arrow[r, "(4.9.3)"] & R\Psi_f(\mathcal{K}_{X_\eta/\eta})(a)[2a] \end{tikzcd} \tag{4.9.4}$$

by the naturality of the nearby cycle functor. By [19, (3.67)], we have a natural map

$$\theta_f : R\Psi_f(\mathcal{K}_{X_\eta/\eta}) \longrightarrow \mathcal{K}_{X_s/s}. \tag{4.9.5}$$

Combining the definition of specialization in [19, (3.68), (3.69)] with (4.9.4), we have

$$\mathrm{sp}^{\text{ét}}(j_X^*\gamma) = \theta_f(a)[2a] \circ R\Psi_f(j_X^*\gamma) \circ (4.9.2) = \theta_f(a)[2a] \circ (4.9.3) \circ i_X^*\gamma. \tag{4.9.6}$$

Since f is smooth, local acyclicity and purity imply $\theta_f(a)[2a] \circ (4.9.3)$ is precisely the isomorphism

$$i_X^*\mathcal{K}_{X/S}(a)[2a] \xrightarrow{\sim} \mathcal{K}_{X_s/s}(a)[2a], \tag{4.9.7}$$

hence the right hand side of (4.9.6) is exactly $i_X^*\gamma$.

□

4.10. We use the following result of SGA $4\frac{1}{2}$ [8]. Assume X' is smooth over S of pure relative dimension N , a closed subscheme $Z' \subset X'$ has fiber codimension at least c and $\mathcal{K}$ is a bounded coherent perfect $\mathcal{O}_{X'}$-complex supported on Z' and of finite Tor-dimension over S. Then [8, Cycle, 2.3.3] constructs a general cycle class

$$\mathrm{cl}_S(Z', \mathcal{K}) \in H^{2c}_{Z'}(X', \Lambda(c)). \tag{4.10.1}$$

Its formation commutes with arbitrary base change, and when S is a field, it agrees with the usual supported cycle class and factors through the Chow group. See [8, Cycle, Lemme 2.3.7, 2.2.10].

4.11. We now prove Theorem 4.8.

Proof of Theorem 4.8. Both paths in (4.8.1) are homomorphisms on Chow groups, so it is enough to treat a prime cycle $[W_\eta]$ supported on Z.

Let $W \subset Z$ be the scheme-theoretic closure of W_η which is integral. Multiplication by u is injective in every affine coordinate ring of W, so W is torsion-free over $\operatorname{Spec} R$, hence flat over S by [16, Tag 0539]. If the special fiber W_s is nonempty, flatness implies $\dim W_s = r$. Set $c = d - r$, the codimension. We have

$$\mathrm{sp}^{\mathrm{cyc}}[W_\eta] = \sum_{\xi \in (W_s)^{(0)}} m_\xi [\overline{\{\xi\}}], \qquad m_\xi = \mathrm{length}_{\mathcal{O}_{W,\xi}}(\mathcal{O}_{W,\xi}/u) \tag{4.11.1}$$

by the previous description of $\mathrm{sp}^{\mathrm{cyc}}([V])$ of a horizontal integral subscheme V. If W_s is empty, the right side of (4.11.1) is zero.

The scheme X is regular and Noetherian because S is a regular trait and f is smooth of finite type; see [16, Tags 0H7S and 0FDC]. Hence the coherent module $\mathcal{O}_{\overline{W}}$ is perfect as an $\mathcal{O}_X$-complex by [16, Tag 0FDC]. Moreover, flatness of W over S implies $\mathcal{O}_{\overline{W}}$ is finite Tor-dimension over S. Thus by [8, Cycle, 2.3.3] we may form

$$\mathrm{cl}_S(W, \mathcal{O}_W) \in H^{2c}_W(X, \Lambda(c)) \simeq H^{-2r}_W(X, \mathcal{K}_{X/S}(-r)). \tag{4.11.2}$$

On the generic fiber, the length of $\mathcal{O}_{W_\eta}$ is one. On the special fiber, flatness gives $Li_X^*\mathcal{O}_W = \mathcal{O}_{W_s}$, whose generic length at ξ is the integer m_ξ in (4.11.1). Therefore by (4.11.1) and [8, Cycle, Lemme 2.3.7, Théorème 2.3.8(i)], we have

$$j_X^*\mathrm{cl}_S(W,\mathcal{O}_W) = \mathrm{cl}_\eta[W_\eta], \tag{4.11.3}$$

$$i_X^*\mathrm{cl}_S(W,\mathcal{O}_W) = \sum_\xi m_\xi \mathrm{cl}_s[\overline{\{\xi\}}] = \mathrm{cl}_s\big(\mathrm{sp}^{\mathrm{cyc}}[W_\eta]\big). \tag{4.11.4}$$

Applying Lemma 4.9, we have

$$\mathrm{sp}^{\text{ét}}\,\mathrm{cl}_\eta[W_\eta] = \mathrm{sp}^{\text{ét}}(j_X^*\mathrm{cl}_S(W,\mathcal{O}_W)) = i_X^*\mathrm{cl}_S(W,\mathcal{O}_W) = \mathrm{cl}_s\big(\mathrm{sp}^{\mathrm{cyc}}[W_\eta]\big). \tag{4.11.5}$$

This proves the commutativity of (4.8.1) on prime cycles. Fulton [10, §20.3] proves that $\mathrm{sp}^{\mathrm{cyc}}$ respects rational equivalence, and the supported cycle class map factors through Chow groups by [8, Cycle, §2.2]. The equality therefore descends to $\mathrm{CH}_r(Z_\eta)$. □

Corollary 4.12. *In the notation of Lemma 4.5, suppose in addition that S is a strictly henselian trait and $B \to S$ is smooth of pure relative dimension d. Put $0 \leq q = m - r \leq d$, then in $H^{-2q}_{Z_s}(B_s, \mathcal{K}_{B_s/s}(-q)) \simeq H^{2(d-q)}_{Z_s}(B_s, \Lambda(d-q))$ we have*

$$\mathrm{sp}^{\text{ét}}\,\mathrm{cl}_\eta(\sigma_\eta^!\gamma_\eta) = \mathrm{cl}_s(\sigma_s^!\gamma_s). \tag{4.12.1}$$

More generally, let

$$\widetilde{\alpha} \in Z_{m+1}(C/S) \tag{4.12.2}$$

be any cycle, denote its class in $\mathrm{CH}_{m+1}(C/S)$ by γ, and set

$$\alpha_\eta := j_C^*\gamma \in \mathrm{CH}_m(C_\eta), \qquad \alpha_s := i_C^!\gamma \in \mathrm{CH}_m(C_s). \tag{4.12.3}$$

Then (4.12.1) takes the form

$$\mathrm{sp}^{\text{ét}}\,\mathrm{cl}_\eta\left(\sigma_\eta^!\alpha_\eta\right) = \mathrm{cl}_s\left(\sigma_s^!\alpha_s\right). \tag{4.12.4}$$

Proof. Apply Theorem 4.8 to $\sigma_\eta^!\gamma_\eta$ we have

$$\mathrm{sp}^{\text{ét}}\,\mathrm{cl}_\eta(\sigma_\eta^!\gamma_\eta) = \mathrm{cl}_s\left(\mathrm{sp}^{\mathrm{cyc}}(\sigma_\eta^!\gamma_\eta)\right). \tag{4.12.5}$$

Now use (4.5.3) we get the desired conclusion. □

Proposition 4.13. *For a separated scheme X of finite type over k, the following conditions are equivalent:*

(1) *there is a finite-dimensional subspace $V \subset \Gamma(X, \mathcal{O}_X)$ such that*

$$V \otimes_k \mathcal{O}_X \longrightarrow \Omega^1_{X/k}, \qquad v \otimes g \longmapsto g \cdot dv, \tag{4.13.1}$$

is surjective.

(2) *X is quasi-affine.*

Proof. First assume condition (2). By [16, Tag 04II], there is a closed immersion $i : X \to U$, where U is an open subscheme of $\mathbb{A}^n_k$. Therefore, the natural homomorphism

$$i^*\Omega^1_{U/k} \to \Omega^1_{X/k} \tag{4.13.2}$$

is a surjection. The consequence follows from

$$i^*\Omega^1_{U/k} \simeq \bigoplus_{j=1}^n \mathcal{O}_X dx_j. \tag{4.13.3}$$

Now we show that (1) implies (2). Let V be the subspace of $\Gamma(X, \mathcal{O}_X)$ satisfying the condition, and fix a basis $g_1, \dots, g_N$. Put

$$g = (g_1, \dots, g_N) : X \to \mathbb{A}^N_k. \tag{4.13.4}$$

In the cotangent sequence

$$g^*\Omega^1_{\mathbb{A}^N_k/k} \longrightarrow \Omega^1_{X/k} \longrightarrow \Omega^1_{X/\mathbb{A}^N_k} \longrightarrow 0, \tag{4.13.5}$$

the first arrow is $dx_i \mapsto dg_i$, where $x_1, \cdots, x_N$ are the coordinates of $\mathbb{A}^N_k$. The surjectivity condition hence implies $\Omega^1_{X/\mathbb{A}^N_k} = 0$ and g is unramified. An unramified morphism is locally quasi-finite; because g is also quasi-compact, it is quasi-finite. See [16, Tags 02V5 and 01TJ].

The diagonal map $X \to X \times_{\mathbb{A}^N_k} X$ is a base change of the diagonal $X \to X \times_k X$; hence g is separated. We may apply Zariski's main theorem to get the factorization

$$X \xrightarrow{j} \overline{X} \xrightarrow{\overline{g}} \mathbb{A}^N_k, \tag{4.13.6}$$

where j is an open immersion and $\overline{g}$ is finite. Hence $\overline{X}$ is affine, and its open subscheme X is quasi-affine.

□

4.14. In the following we use the notations of Theorem 4.2. Let V be a subspace of $\Gamma(X, \mathcal{O}_X)$ obtained by Proposition 4.13, with $N = \dim V$, $\mathbb{V} := \operatorname{Spec}(\operatorname{Sym}_k V^\vee) \simeq \mathbb{A}^N_k$. Define

$$a : X \times_k \mathbb{V} \longrightarrow T^*X, \qquad (x, g) \longmapsto df_x + dg_x. \tag{4.14.1}$$

Lemma 4.15. *There is a dense open subset $\mathbb{V}^\circ \subset \mathbb{V}$ such that $(df + dg)^{-1}(SS(\mathcal{F}))$ is finite for every $g \in \mathbb{V}^\circ(k)$. One may choose $g \in \mathbb{V}(k)$ such that the generic point of the line $k \cdot g$ belongs to $\mathbb{V}^\circ$.*

Proof. The map a lies over the identity of X, and its vertical differential is calculated by the surjective evaluation $V \to T^*_x X$, $g \mapsto dg_x$. Hence a is smooth of relative dimension $N - d$. Since each component of $SS(\mathcal{F})$ has dimension d, every component of

$$Q = a^{-1}(SS(\mathcal{F})) \subset X \times \mathbb{V} \tag{4.15.1}$$

has dimension N.

Let $q : Q \to \mathbb{V}$ be the projection, let $Q_1, \cdots, Q_m$ be the irreducible components of Q. Put

$$D := \bigcup_{\overline{q(Q_i)} \subsetneq \mathbb{V}} \overline{q(Q_i)} \subsetneq \mathbb{V}, \tag{4.15.2}$$

$U_D := \mathbb{V} \setminus D$ is the complement. Then for any $v \in U_D$, and Q_i with $\overline{q(Q_i)} \subsetneq \mathbb{V}$, we have $(Q_i)_v = \varnothing$.

For Q_j such that Q_j dominates $\mathbb{V}$, i.e. $\overline{q(Q_j)} = \mathbb{V}$, we have

$$\dim(Q_j)_\eta = \dim Q_j - \dim \mathbb{V} = 0 \tag{4.15.3}$$

where η is the generic point of $\mathbb{V}$. By a generic fiber dimension result [16, Tag 05F7], $\dim(Q_j)_u = 0$ for all u in an open subscheme U_j of $\mathbb{V}$. Therefore,

$$\mathbb{V}^\circ := \Big(\bigcap_{Q_j \text{ dominates } \mathbb{V}} U_j\Big) \cap U_D \tag{4.15.4}$$

has the required property.

Choose $v \in \mathbb{V}^\circ(k)$. The line $k \cdot v$ is not contained in $\mathbb{V} \setminus \mathbb{V}^\circ$, because it contains v; hence its generic point belongs to $\mathbb{V}^\circ$. Now $g = v$ verifies the demand of the lemma. □

4.16. Choose g as in Lemma 4.15. Let R be the strict henselization of $k[u]_{(u)}$, set $S = \operatorname{Spec} R$, write $K = \operatorname{Frac}(R)$, and put $\eta^{\mathrm{perf}} = \operatorname{Spec} K^{\mathrm{perf}}$. The natural inclusion $k \to R$ gives $S \to \operatorname{Spec} k$. Let $\eta = \operatorname{Spec} K$, $s = \operatorname{Spec} k$ be the generic and special point of S, respectively. Define

$$\mathcal{X} = X \times_k S, \qquad \mathcal{Y} = \mathbb{A}^1_S, \qquad \widetilde{\mathcal{F}} = \mathrm{pr}_X^* \mathcal{F}, \tag{4.16.1}$$

$$\phi : \mathcal{X} \to \mathcal{Y} : \phi(x, u) = f(x) + ug(x), \tag{4.16.2}$$

where $\mathrm{pr}_X : \mathcal{X} \to X$ is the projection. The differential $d\phi$ gives a section

$$\sigma = d\phi = df + u\, dg : \mathcal{X} \longrightarrow T^*(\mathcal{X}/S) = T^*X \times_k S. \tag{4.16.3}$$

Put $\mathcal{W} = \sigma^{-1}(SS(\mathcal{F}) \times_k S)$, then by Lemma 4.15,

$$\mathcal{W}_s = Z = Z_{X,f}, \qquad \mathcal{W}_\eta \text{ is finite over } K, \tag{4.16.4}$$

and hence $\mathcal{W}_{\eta^{\mathrm{perf}}}$ is finite over K^{perf}.

For $t \in \{\eta, s\}$, ϕ_t is $(SS(\mathcal{F}) \times_k S)_t$-transversal outside $\mathcal{W}_t$. For $t = s$, [14, Lemma 4.2(5)] implies that ϕ_s is universally locally acyclic relatively to $\widetilde{\mathcal{F}}_s$ outside $\mathcal{W}_s$.

The projection

$$\mathrm{pr}_S : \mathcal{X} = X \times_k S \to S \tag{4.16.5}$$

is universally locally acyclic relatively to $\widetilde{\mathcal{F}}$, and

$$R\Psi_{\mathrm{pr}_S}(\widetilde{\mathcal{F}}_\eta) \simeq \widetilde{\mathcal{F}}_s \simeq \mathcal{F}. \tag{4.16.6}$$

4.17. Retain the notation of §4.16. Initially, the generic fiber is over K which is not perfect in general; however, the construction of characteristic cycle in [14] is over a perfect field. In the following, we build a descent result to pass to the perfect closure of K.

Lemma 4.18. *Assume that k is algebraically closed. Let X be a smooth and purely d-dimensional scheme over k, let $\mathcal{F} \in D_{\mathrm{ctf}}(X, \Lambda)$, and let K^{perf} be the perfect closure of K. Then we have*

$$SS(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = SS(\mathcal{F}) \times_k \eta^{\mathrm{perf}}, \qquad CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}}, \tag{4.18.1}$$

where the second term is the flat pullback of cycle.

Proof. Under the canonical identification

$$T^*(\mathcal{X}_{\eta^{\mathrm{perf}}}/K^{\mathrm{perf}}) \simeq T^*(X/k) \times_k \eta^{\mathrm{perf}}, \tag{4.18.2}$$

the complex $\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}$ is the scalar extension of $\mathcal{F}$. Hence by [4, Theorem 1.4(iii)] we have

$$SS(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = SS(\mathcal{F}) \times_k \eta^{\mathrm{perf}}. \tag{4.18.3}$$

Put

$$C = SS(\mathcal{F}) = \bigcup_\alpha C_\alpha, \qquad CC(\mathcal{F}) = \sum_\alpha m_\alpha [C_\alpha], \tag{4.18.4}$$

where the C_α are all the irreducible components of C, allowing $m_\alpha = 0$, and set

$$C_{\mathrm{perf}} = C \times_k \eta^{\mathrm{perf}}, \qquad B = CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}}. \tag{4.18.5}$$

Every irreducible component of C_{perf} has dimension d, and B is supported on C_{perf}. It suffices to verify the Milnor formula for B.

Let

$$j : U \longrightarrow \mathcal{X}_{\eta^{\mathrm{perf}}}, \qquad f : U \longrightarrow Y \tag{4.18.6}$$

be a test pair with an isolated characteristic point, i.e., j is étale, Y is a smooth curve over K^{perf}, and $u_0 \in U$ is a closed point, with a geometric point $\bar{u}$ above it, which is at most an isolated j^*C_{perf}-characteristic point of f.

If f is j^*C_{perf}-transversal at u_0, then we may shrink U around u_0. The local intersection number is zero and [14, Lemma 4.2(5)] gives

$$R\Phi_{\bar{u}}(j^*\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, f) = 0. \tag{4.18.7}$$

Thus the Milnor formula holds in this case.

Assume henceforth that u_0 is isolated characteristic point. In this case, we may assume $Y = \mathbb{A}^1_{K^{\mathrm{perf}}}$ by shrinking around u_0 and passing to an étale cover. Write

$$df : U \longrightarrow T^*(U/K^{\mathrm{perf}}) \tag{4.18.8}$$

for the section defined by the relative differential of f. Since j is étale, we may regard j^*C_{perf} as a closed conical subset of $T^*(U/K^{\mathrm{perf}})$, endowed with its reduced scheme structure. After shrinking U once more, assume that the scheme-theoretic characteristic locus

$$Z_{\mathrm{perf}} := U \times_{df, T^*(U/K^{\mathrm{perf}})} j^*C_{\mathrm{perf}} \tag{4.18.9}$$

is finite over K^{perf}.

The field K is algebraic over $k(u)$, since it is the fraction field of the strict henselization of $k[u]_{(u)}$, and K^{perf}/K is purely inseparable. Hence $K^{\mathrm{perf}}/k(u)$ is algebraic. Since all data in the test pair are of finite presentation, they descend to a finitely generated integral $k[u]$-subalgebra $A_0 \subset K^{\mathrm{perf}}$. The extension $\mathrm{Frac}(A_0)/k(u)$ is finite. After localizing A_0, write A for the resulting ring. Since k is perfect, we may choose this localization so that $A \subset K^{\mathrm{perf}}$ is a finitely generated smooth integral k-algebra containing $k[u]$ and carrying the descended data. Put $M = \operatorname{Spec} A, F_0 = \mathrm{Frac}(A)$. Thus A is of finite type over k, and K^{perf}/F_0 is algebraic. After enlarging the model if necessary, we obtain an étale morphism and an M-morphism

$$j_A : U_A \longrightarrow X \times_k M, \qquad f_A : U_A \longrightarrow \mathbb{A}^1_M \tag{4.18.10}$$

whose base change to η^{perf} is the given test. Put

$$h = \mathrm{pr}_X \circ j_A : U_A \to X, \quad \pi = \mathrm{pr}_M \circ j_A : U_A \longrightarrow M, \quad \mathcal{F}_A = j_A^* \,\mathrm{pr}_X^* \,\mathcal{F}, \tag{4.18.11}$$

and, using the cotangent-bundle isomorphism induced by j_A, put

$$C_A = U_A \times_X C \subset T^*(U_A/M), \tag{4.18.12}$$

$$B_A = j_A^*(CC(\mathcal{F}) \times_k M) = \sum_\alpha m_\alpha [U_A \times_X C_\alpha]. \tag{4.18.13}$$

Let $df_A : U_A \longrightarrow T^*(U_A/M)$ be the section defined by the relative differential of f_A, and define

$$Z_A = U_A \times_{df_A, T^*(U_A/M)} C_A. \tag{4.18.14}$$

We have $(Z_A)_{\eta^{\mathrm{perf}}} = Z_{\mathrm{perf}}$.

The generic fiber $(Z_A)_{F_0}$ is finite, since it becomes the finite scheme Z_{perf} after the faithfully flat extension $F_0 \subset K^{\mathrm{perf}}$. The open quasi-finite locus of $Z_A \to M$ therefore contains the generic fiber. Its complement has only finitely many irreducible components, none of which dominates M. After removing from M the closures of their images, we may assume that $Z_A \longrightarrow M$ is quasi-finite.

Since $j_A = (h, \pi)$ is étale, the morphism h is smooth. Hence h is C-transversal and $SS(\mathcal{F}_A) = h^\circ C$ by [14, Lemma 3.4(1) and Corollary 4.5(2)]. Under the cotangent decomposition induced by j_A, this equality becomes

$$T^*(U_A/k) \simeq T^*(U_A/M) \oplus \pi^*T^*(M/k), \qquad h^\circ C = C_A \oplus 0. \tag{4.18.15}$$

For the M-morphism f_A, the composition of its cotangent map

$$df_A : T^*(\mathbb{A}^1_M/k) \times_{\mathbb{A}^1_M} U_A \to T^*(U_A/k) \tag{4.18.16}$$

with the projection $T^*(U_A/k) \longrightarrow T^*(U_A/M)$ sends $(a\,dt, \beta)$ to $a\,df_A$, while covectors pulled back from M map identically to the second summand $\pi^*T^*(M/k)$. Consequently, by the definition of transversality and the conicality of C_A, the non-C-transversal locus of (h, f_A) is precisely the underlying set of $(df_A)^{-1}(C_A) = Z_A$. Thus (h, f_A) is C-transversal on $U_A \setminus Z_A$. Since the cotangent map of π has image in the second summand $\pi^*T^*(M/k)$, which meets $C_A \oplus 0$ only in the zero-section, the pair (h, π) is C-transversal everywhere.

For a geometric point $\bar{z}$ of Z_A, with image $\bar{s}$ in M, define

$$\begin{aligned} I(\bar{z}) &= (B_{A,\bar{s}}, df_{A,\bar{s}})_{T^*(U_{A,\bar{s}}/\bar{s}),\bar{z}}, \\ \phi(\bar{z}) &= -\operatorname{dimtot} R\Phi_{\bar{z}}(\mathcal{F}_{A,\bar{s}}, f_{A,\bar{s}}). \end{aligned} \tag{4.18.17}$$

For every α, set

$$\mathcal{A}_{\alpha,A} = \mathcal{O}_{U_A \times_X C_\alpha} \otimes^{\mathbf{L}}_{\mathcal{O}_{T^*(U_A/M)}} \mathcal{O}_{U_A}, \tag{4.18.18}$$

where the last module structure is defined by the section df_A. Since M is smooth over k and U_A is smooth over M, the scheme $T^*(U_A/M)$ is regular. Hence $\mathcal{O}_{U_A \times_X C_\alpha}$ is perfect on it, and $\mathcal{A}_{\alpha,A}$ is perfect on U_A. Since $U_A \to M$ is smooth, $\mathcal{A}_{\alpha,A}$ has finite Tor-dimension over M. Moreover,

$$U_A \times_X C_\alpha = U_A \times_{X \times_k M} (C_\alpha \times_k M) \tag{4.18.19}$$

is étale over $C_\alpha \times_k M$, and therefore is flat over M; the scheme U_A is also flat over M. By derived base change we have

$$(\mathcal{A}_{\alpha,A})_{\overline{s}} = Li^*_{\overline{s}} \mathcal{A}_{\alpha,A} \simeq \mathcal{O}_{U_{A,\overline{s}} \times_X C_\alpha} \otimes^L_{\mathcal{O}_{T^*(U_{A,\overline{s}}/\overline{s})}, df_{A,\overline{s}}} \mathcal{O}_{U_{A,\overline{s}}} \tag{4.18.20}$$

where $\overline{s} \to M$ is any geometric point. Thus $(\mathcal{A}_{\alpha,A})_{\overline{s}}$ is the derived local intersection of $U_{A,\overline{s}} \times_X C_\alpha$ and $df_{A,\overline{s}}(U_{A,\overline{s}})$ in $T^*(U_{A,\overline{s}}/\overline{s})$. Its local Euler characteristic is the corresponding isolated intersection multiplicity in the definition of the function I. The cohomology of $\mathcal{A}_{\alpha,A}$ is supported on

$$(U_A \times_X C_\alpha) \cap df_A(U_A) \subset Z_A, \tag{4.18.21}$$

hence on a closed subscheme quasi-finite over M. Therefore [14, Lemma 2.3 and the proof of Proposition 5.8], followed by summing with the coefficients m_α, shows that I is constructible and flat over M in the sense of [14, Definition 2.1].

The transversality of (h, π) and of (h, f_A) off Z_A, together with [14, Lemma 4.2(5)], shows that π is universally locally acyclic relatively to $\mathcal{F}_A$ and that f_A is universally locally acyclic relatively to $\mathcal{F}_A$ on $U_A \setminus Z_A$. By [14, Proposition 2.16] we have ϕ is also constructible and flat over M since $\mathbb{A}^1_M \to M$ is a smooth relative curve and $Z_A \to M$ is quasi-finite.

Let t be a closed point of M. Since A is of finite type over the algebraically closed field k, one has $\kappa(t) = k$, and $h_t : U_{A,t} \longrightarrow X$ is étale. By [14, Lemma 5.11(2)], we have

$$B_{A,t} = h_t^* CC(\mathcal{F}) = CC(h_t^* \mathcal{F}) = CC(\mathcal{F}_{A,t}). \tag{4.18.22}$$

The fiber $Z_{A,t}$ is finite. Saito's Milnor formula [14, Theorem 5.9] therefore gives

$$I(\bar{z}) = \phi(\bar{z}) \tag{4.18.23}$$

for every geometric point $\bar{z}$ above every closed point $t \in M$.

Notice that $I - \phi$ is also a constructible function. On every irreducible component Z' of Z_A, constructibility gives a dense open subset V on which $I - \phi$ equals its value at the geometric generic point of Z'. Since Z' is of finite type over the algebraically closed field k, it is Jacobson, and V contains a closed k-point z; its image in M, z_M, is also closed. Hence regard $\bar{z}$ as a geometric point of Z_A lying above z_M, by (4.18.23) we have $I(\overline{z}) = \phi(\overline{z})$ which implies $I(z) = \phi(z)$. Thus $I - \phi$

vanishes at the generic point of every irreducible component of Z_A. Saito [14, Lemma 2.2(3)] now gives $I - \phi = 0$ on all geometric points of Z_A.

Choose an algebraic closure Ω of K^{perf} together with an embedding $\kappa(u_0) \hookrightarrow \Omega$ denoted by $\bar{u}$. Since K^{perf}/F_0 is algebraic, Ω is also an algebraic closure of F_0, so $\bar{u}$ defines a geometric point of Z_A above the geometric generic point of M. Moreover, $\kappa(u_0)/K^{\mathrm{perf}}$ is finite separable because K^{perf} is perfect. Flat base change therefore preserves the local intersection multiplicity at the chosen geometric point, hence we have

$$I(\bar{u}) = (j^*B, df)_{T^*(U/K^{\mathrm{perf}}),u_0}. \tag{4.18.24}$$

Since $Z_A \to M$ is quasi-finite, the induced morphism $Z_A \to \mathbb{A}^1_M$ is also quasi-finite. By the universal local acyclicity of f_A outside Z_A and [14, Proposition 2.8(1)], we may identifies the vanishing-cycle complex defining $\phi(\bar{u})$ with $R\Phi_{\overline{u}}(j^*\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, f)$. Consequently,

$$\phi(\bar{u}) = -\operatorname{dimtot} R\Phi_{\bar{u}}(j^*\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, f). \tag{4.18.25}$$

Evaluating $I - \phi = 0$ at $\bar{u}$ yields

$$(j^*B, df)_{T^*(U/K^{\mathrm{perf}}),u_0} = -\operatorname{dimtot} R\Phi_{\bar{u}}(j^*\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, f). \tag{4.18.26}$$

Thus B satisfies the Milnor formula for every at most isolated C_{perf}-characteristic test. Since $C_{\mathrm{perf}} = SS(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}})$, by the uniqueness in [14, Theorem 5.9] we have

$$CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}}. \tag{4.18.27}$$

This proves (4.18.1).

□

4.19. We now verify the local-acyclicity condition on the ordinary generic fiber. By (4.18.1),

$$SS(\mathcal{F}) \times_k \eta^{\mathrm{perf}} = SS(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}). \tag{4.19.1}$$

Since $\phi_{\eta^{\mathrm{perf}}}$ is $SS(\mathcal{F})_{\eta^{\mathrm{perf}}}$-transversal outside $\mathcal{W}_{\eta^{\mathrm{perf}}}$, [14, Lemma 4.2(5)] implies that $\phi_{\eta^{\mathrm{perf}}}$ is universally locally acyclic relatively to $\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}$ outside $\mathcal{W}_{\eta^{\mathrm{perf}}}$.

The morphism $\eta^{\mathrm{perf}} = \operatorname{Spec} K^{\mathrm{perf}} \to \operatorname{Spec} K = \eta$ is a universal homeomorphism. Every base change is again a universal homeomorphism, and the equivalences of étale topoi in [16, Tag 03SI] identify the local-acyclicity criterion of [8, Th. finitude, Définition 2.12]. Hence universal local acyclicity descends along this extension, i.e.,

$$\phi_\eta : \mathcal{X}_\eta \longrightarrow \mathcal{Y}_\eta \tag{4.19.2}$$

is universally locally acyclic relatively to $\widetilde{\mathcal{F}}_\eta$ outside $\mathcal{W}_\eta$. By a slight abuse of notation, we denote both the specialization map on $H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta})$ and $H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta})$ by $\mathrm{sp}^{\text{ét}}$. The universal local acyclicity of $\mathrm{pr}_S : \mathcal{X} \to S$ gives

$$R\Psi_{\mathrm{pr}_S}(\widetilde{\mathcal{F}}_\eta) \simeq \widetilde{\mathcal{F}}_s \simeq \mathcal{F}, \tag{4.19.3}$$

therefore Proposition 3.3(6) gives

$$\mathrm{sp}^{\text{ét}}\, \widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) = \widetilde{C}^{Z}_{X/\mathbb{A}^1_k/k}(\mathcal{F}). \tag{4.19.4}$$

4.20. Recall that we have the section $\sigma = d\phi = df + u\,dg$. Combining Lemma 4.18 with flat base change for the refined intersection, we have

$$\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta) \times_\eta \eta^{\mathrm{perf}} = \sigma_{\eta^{\mathrm{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}}) = \sigma_{\eta^{\mathrm{perf}}}^! CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}). \tag{4.20.1}$$

Lemma 4.21. *With the preceding notation, we have*

$$\widetilde{C}^{\mathcal{W}_{\eta^{\mathrm{perf}}}}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = \widetilde{\mathrm{cl}}\Big(\sigma_{\eta^{\mathrm{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}})\Big). \tag{4.21.1}$$

Proof. Recall that under our construction, $\mathcal{W}_{\eta^{\mathrm{perf}}}$ is finite over η^{perf}. For each closed point $x \in |\mathcal{W}_{\eta^{\mathrm{perf}}}|$, choose an open neighbourhood U_x containing no other point of $|\mathcal{W}_{\eta^{\mathrm{perf}}}|$. The étale locality and the cohomological Milnor formula of non-acyclicity class [19, Proposition 4.6 and Theorem 6.1] give

$$C^{\{x\}}_{U_x/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}|_{U_x}) = -\operatorname{dimtot} R\Phi_{\bar{x}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, \phi_{\eta^{\mathrm{perf}}}) \quad \text{in } H^0_x(U_x, \mathcal{K}_{U_x/\eta^{\mathrm{perf}}}) \simeq \Lambda. \tag{4.21.2}$$

Saito's isolated characteristic-cycle formula [14, Theorem 5.9] says, for the same point,

$$\begin{aligned} &\mathrm{mult}_x\Big(\sigma_{\eta^{\mathrm{perf}}}^! CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}})\Big) \\ &\quad = (CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}), d\phi_{\eta^{\mathrm{perf}}})_{T^*(\mathcal{X}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}),x} = -\operatorname{dimtot} R\Phi_{\bar{x}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}, \phi_{\eta^{\mathrm{perf}}}) \quad \text{in } \mathbf{Z}. \end{aligned} \tag{4.21.3}$$

Since supported cohomology on a finite set decomposes as the direct sum of its point-supported groups, respects the cycle class map cl, summing over x gives

$$\widetilde{C}^{\mathcal{W}_{\eta^{\mathrm{perf}}}}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}) = \widetilde{\mathrm{cl}}(\sigma_{\eta^{\mathrm{perf}}}^! CC(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}})). \tag{4.21.4}$$

Applying (4.20.1) we get the desired result. ∎

Lemma 4.22. *Let $q_X : \mathcal{X}_{\eta^{\mathrm{perf}}} \to \mathcal{X}_\eta$ and $q_Y : \mathcal{Y}_{\eta^{\mathrm{perf}}} \to \mathcal{Y}_\eta$ be the projections. Then there is a canonical isomorphism*

$$q_X^* \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}, \tag{4.22.1}$$

compatible with the maps from the absolute dualizing complexes. Consequently

$$q_X^* : H^0_{\mathcal{W}_\eta}(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}) \xrightarrow{\sim} H^0_{\mathcal{W}_{\eta^{\mathrm{perf}}}}(\mathcal{X}_{\eta^{\mathrm{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}). \tag{4.22.2}$$

Proof. The maps q_X and q_Y are universal homeomorphisms, hence q_X^* and q_Y^* are equivalences on the relevant derived categories of constructible sheaves by [3, Exposé VIII, Théorème 1.1]. For the Cartesian square defined by $\phi_\eta : \mathcal{X}_\eta \to \mathcal{Y}_\eta$, base change for lower shriek gives

$$q_Y^* R\phi_{\eta!} \xrightarrow{\sim} R\phi_{\eta^{\mathrm{perf}}!} q_X^*, \tag{4.22.3}$$

by [16, Tag 0F7L]. Moreover, the two equivalences yield a canonical base change isomorphism

$$q_X^* R\phi_\eta^! \xrightarrow{\sim} R\phi_{\eta^{\mathrm{perf}}}^! q_Y^*. \tag{4.22.4}$$

Hence by definition we have

$$q_X^* \mathcal{K}_{\mathcal{X}_\eta/\eta} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}, \qquad q_X^* \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta} \xrightarrow{\sim} \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}}, \tag{4.22.5}$$

and the consequence follows from the distinguished triangle (3.2.3). ∎

Lemma 4.23. *Retain the preceding notation. We have*

$$\widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) = \widetilde{\mathrm{cl}}\Big(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)\Big). \tag{4.23.1}$$

Proof. Let q_X and q_Y be as before. The distinguished triangle (3.2.3) gives

$$\rho_s : H^0(X, \mathcal{K}_{X/k}) \to H^0(X, \mathcal{K}_{X/\mathbb{A}^1_k/k}), \qquad \rho_\eta : H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta}) \to H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}), \tag{4.23.2}$$

and similarly we define $\rho_{\eta^{\mathrm{perf}}}$. Thus

$$\widetilde{\mathrm{cl}}_\eta = \rho_\eta \circ \mathrm{cl}_\eta, \qquad \widetilde{\mathrm{cl}}_{\eta^{\mathrm{perf}}} = \rho_{\eta^{\mathrm{perf}}} \circ \mathrm{cl}_{\eta^{\mathrm{perf}}}. \tag{4.23.3}$$

We compare the pullbacks of the two sides of (4.23.1). First, the pullback formula for the non-acyclicity class (3.3.2) gives

$$q_X^* \widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) = \widetilde{C}^{\mathcal{W}_{\eta^{\mathrm{perf}}}}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}(\widetilde{\mathcal{F}}_{\eta^{\mathrm{perf}}}). \tag{4.23.4}$$

On the cycle side, compatibility of the supported cycle-class map with scalar extension [8, Cycle, Lemme 2.3.7] gives

$$q_X^* \mathrm{cl}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) = \mathrm{cl}_{\eta^{\mathrm{perf}}}(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta) \times_\eta \eta^{\mathrm{perf}}). \tag{4.23.5}$$

Moreover, the naturality of morphisms involved gives the commutative square

$$\begin{CD} H^0_{\mathcal{W}_\eta}(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta}) @>{\rho_\eta}>> H^0_{\mathcal{W}_\eta}(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}) \\ @V{q_X^*}VV @VV{q_X^*}V \\ H^0_{\mathcal{W}_{\eta^{\mathrm{perf}}}}(\mathcal{X}_{\eta^{\mathrm{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}) @>{\rho_{\eta^{\mathrm{perf}}}}>> H^0_{\mathcal{W}_{\eta^{\mathrm{perf}}}}(\mathcal{X}_{\eta^{\mathrm{perf}}}, \mathcal{K}_{\mathcal{X}_{\eta^{\mathrm{perf}}}/\mathcal{Y}_{\eta^{\mathrm{perf}}}/\eta^{\mathrm{perf}}}). \end{CD} \tag{4.23.6}$$

Consequently, using (4.20.1), we have

$$\begin{aligned} q_X^* \widetilde{\mathrm{cl}}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)) &= \widetilde{\mathrm{cl}}_{\eta^{\mathrm{perf}}}(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta) \times_\eta \eta^{\mathrm{perf}}) \\ &= \widetilde{\mathrm{cl}}_{\eta^{\mathrm{perf}}}(\sigma_{\eta^{\mathrm{perf}}}^!(CC(\mathcal{F}) \times_k \eta^{\mathrm{perf}})). \end{aligned} \tag{4.23.7}$$

By (4.21.1), the right-hand sides of (4.23.4) and (4.23.7) are equal, hence

$$q_X^* \widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) = q_X^* \widetilde{\mathrm{cl}}_\eta(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)). \tag{4.23.8}$$

Finally, by the previous lemma 4.22, the equality descends uniquely to η, and we obtain

$$\widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) = \widetilde{\mathrm{cl}}_\eta\Big(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)\Big), \tag{4.23.9}$$

which is (4.23.1). $\square$

Proof of Theorem 4.2. Recall

$$\rho_s : H^0(X, \mathcal{K}_{X/k}) \to H^0(X, \mathcal{K}_{X/\mathbb{A}^1_k/k}), \qquad \rho_\eta : H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\eta}) \to H^0(\mathcal{X}_\eta, \mathcal{K}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}) \tag{4.23.10}$$

and by definition we have $\widetilde{\mathrm{cl}}_s = \rho_s \circ \mathrm{cl}_s$, $\widetilde{\mathrm{cl}}_\eta = \rho_\eta \circ \mathrm{cl}_\eta$. By [19, (4.47)], we have

$$\mathrm{sp}^{\text{ét}} \circ \rho_\eta = \rho_s \circ \mathrm{sp}^{\text{ét}}. \tag{4.23.11}$$

Notice that every component of the cycle $CC(\mathcal{F}) \times_k [S] \in \mathrm{CH}_{d+1}(SS(\mathcal{F}) \times_k S/S)$ is flat over S, Corollary 4.12 gives

$$\mathrm{sp}^{\text{ét}}\, \mathrm{cl}_\eta\Big(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)\Big) = \mathrm{cl}_s\Big((df)^! CC(\mathcal{F})\Big) \tag{4.23.12}$$

Combining (4.19.4), (4.23.1) and (4.23.12), we have

$$\begin{aligned}\widetilde{C}^{Z}_{X/\mathbb{A}^1/k}(\mathcal{F}) &= \mathrm{sp}^{\text{ét}}\, \widetilde{C}^{\mathcal{W}_\eta}_{\mathcal{X}_\eta/\mathcal{Y}_\eta/\eta}(\widetilde{\mathcal{F}}_\eta) \\ &= \mathrm{sp}^{\text{ét}}\, \rho_\eta \mathrm{cl}\Big(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)\Big) \\ &= \rho_s\, \mathrm{sp}^{\text{ét}}\, \mathrm{cl}\Big(\sigma_\eta^!(CC(\mathcal{F}) \times_k \eta)\Big) \\ &= \rho_s \mathrm{cl}\big((df)^! CC(\mathcal{F})\big).\end{aligned} \tag{4.23.13}$$

This is Theorem 4.2. □

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